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A Flux Reconstruction Approach to High-Order Schemes Including Discontinuous Galerkin Methods

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[Abstract] We introduce a new approach to high-order accuracy for the numerical solution of conservation laws by using flux reconstruction. The approach bridges the two existing approaches of discontinuous Galerkin (DG) and staggered-grid (or spectral difference), and leads to simplified versions of these two methods. It also results in several new schemes with favorable properties. Schemes via the current approach are simple and economical: they solve the conservation laws in differential form and involve only one grid (not two grids that are staggered). In addition, they are conservative. As a result of the new approach, the DG scheme can be formulated using the differential instead of integral form. Another result is a new method, which is similar to the staggered-grid scheme, but more accurate, economical and stable.

I. Introduction

In the field of Computational Fluid Dynamics, low-order methods are less accurate, but generally are robust and reliable; as a result, they are routinely employed in practical calculations. High-order (third and above) methods have the potential of providing accurate solutions at reasonable cost; however, they are more complicated and less robust. The need to improve and refine high-order methods as well as to develop new ones with more favorable properties has attracted the interest of many researchers. The current work is a step in this direction.

For the case of one spatial dimension, in a high-order semi-discretization of conservation laws, the solution can be approximated in each cell (interval) by the data at K points called solution points, where K is a positive integer. These K pieces of data define a polynomial of degree $K-1$, herein called the *solution polynomial*. This polynomial interpolates the data *within* a cell, a feature of the finite-element methods; in contrast, the finite-volume methods employ interpolation across several cells. The solution polynomials collectively form a function, which can be and usually is discontinuous across cell interfaces. At each interface, from the two values to its left and right, a common flux, which is typically the upwind flux, a key feature of the finite-volume methods, is defined. The common or upwind flux allows the data in adjacent cells to interact and its commonality results in a conservative scheme.

The upwind flux can be employed in two different approaches. The first approach, introduced by Reed and Hill for neutron transport equations (1973) and developed for fluid dynamics equations by Cockburn and Shu and others (Cockburn, Karniadakis, and Shu 2000 and the references therein), is the discontinuous Galerkin or DG approach. Here, after multiplying the conservation law with a test function, integrating over a cell, and applying an integration by parts, the upwind fluxes are employed to evaluate the boundary integral. For the two-dimensional case, the approach can be formulated on triangular as well as quadrilateral meshes. If the mesh is quadrilateral, and the solution points are chosen to be the tensor products of the Gauss points, the DG schemes become highly economical and are called the discontinuous spectral element methods (Kopriva, Woodruff, and Hussaini 2000).

The second approach, introduced by Kopriva and Kolias (1996), solves the differential form of the conservation law as opposed to the integral form, and is referred to as the staggered-grid approach. (Note that this type of staggered grid is completely independent of the type commonly employed in incompressible flows where the location for pressure is staggered relative to that of velocity.) Here, the solution points are the K Chebyshev-Gauss points; the flux points, which include the two cell interfaces, are the $K+1$ Chebyshev-Lobatto points; and these two sets of points are staggered. In each cell, the solution polynomial is of degree $K-1$ and is defined by the values at the solution points. The flux polynomial is of degree K and defined by the upwind flux values at the two interfaces together with the flux values at the $K-1$ interior flux points. The spatial derivatives of the flux polynomial at the

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solution points are employed to update the solution. For the case of triangular meshes, this approach is more involved and results in the spectral difference schemes (Liu, Vinokur, and Wang 2003, 2005).

The exact relationship between the DG and staggered-grid schemes has not been established. The reason, perhaps, is that the two approaches are very different. The DG approach is highly mathematical and provides powerful machinery to obtain the solution. However, the approach is not intuitive; e.g., it is not clear how oscillations are generated near a discontinuity. The staggered-grid approach, on the other hand, is intuitive; e.g., one can easily see how oscillations are generated near a discontinuity by plotting the flux polynomials. Mathematically, however, it remains to be shown whether the schemes using staggered grids are stable and/or optimal.

In addition to the above differences, there is also the following trade-off: the DG scheme is more accurate, but the staggered-grid has a larger CFL limit. (The CFL or Courant, Friedrichs, and Lewy limit is the largest nondimensional time step allowed by stability condition.) Moreover, for scalar conservation laws, but not for the Euler equations, the DG scheme is super-accurate: its order of accuracy is $2K-1$ (Biswas et al. 1994, Adjerid et al. 2002). A scheme using K solution points is said to be super-convergent or super-accurate if its order of accuracy is higher than *K*. The staggered-grid scheme, as will be shown below, is not super-accurate. In general, it is not well understood what makes a scheme super-accurate or whether there is a relation between super-accuracy and CFL limit.

In this paper, we introduce a new flux reconstruction approach, which is considerably more general than that of the staggered grid scheme. The approach establishes the relation between the two existing approaches of DG and staggered-grid and leads to simplified versions of these two methods as well as to several new schemes. Schemes via the new approach are simple and economical: they solve the conservation laws in differential form and involve only one grid, namely the solution points. In addition, they are conservative. As a result of the new approach, the DG scheme can be formulated using the differential instead of integral form. The key ideas in the one-dimensional case are the following. The flux values at the *K* solution points define a polynomial of degree $K-1$. These polynomials collectively form a function, which is *discontinuous* across cell interfaces. Approximating the derivative of the flux by that of this discontinuous function results in bad solutions since such a derivative ignores the neighboring data; in other words, it involves no interaction with the data in adjacent cells. To account for the interaction (or the jumps at interfaces), we first construct a *continuous* flux function, which approximates the discontinuous function in some sense. At each interface, we require this continuous function to take the upwind flux value; thus, continuity across interfaces is assured. In each cell, the continuous flux function must be a polynomial of degree *K* so that its (spatial) derivative is of degree $K-1$, consistent with the degree of the solution polynomial. By considering the difference of the continuous and discontinuous flux functions, it suffices to define a correction function *g*, which approximates the zero function. Focusing on the correction due to the jump only at the left boundary, factoring out the size of the jump, and assuming the cell is the interval $I=[-1,1]$, the problem reduces to defining a polynomial g of degree *K* such that $g(-1)=1$, $g(1)=0$, and g approximates the function 0 in some sense. The condition $g(-1)=1$ deals with the jump at the left interface whereas $g(1)=0$ leaves the right interface value unchanged. Since two conditions are set, g is determined by $K-1$ additional conditions. At each solution point, the derivative of g is then employed to correct that of the *discontinuous* flux function. The correction due to the jump at the right boundary follows by symmetry. Note the following differences compared to the staggered-grid approach: the use of the discontinuous flux function; the separation of the correction at the left interface from that of the right; the introduction of the correction function g; and the use of only one grid. While the best choice for g is likely to be problem dependent, we present several choices and discuss the trade-offs.

The first choice for g results in a simplified version of the DG scheme and can be described as follows. Denote by $\boldsymbol{P}_m$ the space of polynomials of degree *m* or less on I with the integral inner product. If the $K-1$ additional conditions are that the projection of g onto $\boldsymbol{P}_{K-2}$ is 0 (i.e., g approximates the function 0 in the least-squares sense), then g is the right Radau polynomial, and the resulting scheme is the DG method. Alternatively, this scheme can be obtained by defining the continuous flux function in each cell via the two interface upwind fluxes and the $K-1$ conditions that its projection onto $\boldsymbol{P}_{K-2}$ is the same as that of the discontinuous flux function. The current formulation of DG via the correction function g is simple and economical: it uses only derivative formulas at the solution points and involves no quadratures. In fact, it is as economical as the spectral element scheme (Kopriva et al. 2000) with the added advantages of conceptual simplicity and coding simplicity and economy regardless of the choice of solution points.

Kopriva and Kolias' staggered-grid scheme can also be cast in this approach: the $K-1$ additional conditions for g are that g vanishes at the $K-1$ interior Chebyshev-Lobatto (flux) points; that is, g is defined by interpolation rather than projection (least squares). The current version of the scheme is, as in the DG case, simpler than the

original: we deal only with one grid, namely, the solution points, and they can be the Gauss, Lobatto, or equidistant points. (Note that for our version, the name staggered-grid is no longer appropriate.) Via Fourier analysis, we also show that the staggered-grid scheme is mildly unstable. This problem can be remedied by one of the new methods: the scheme (with g defined via interpolation) using the two cell boundaries and the $K-1$ Legendre-Gauss points as flux points is stable. This scheme is accurate to order $2K-2$. Compared to DG, its accuracy is one order lower, but to its advantage, the CFL limit is roughly 3/2 times as large.

Among new members, the scheme using the K Legendre-Lobatto points as solution points, together with a correction function that lumps the derivative corrections to the boundary, is simplest: the interface values are readily available without interpolation, and the derivative corrections only take effect at the corresponding boundary and not at any other solution points. This scheme is also accurate to order $2K-2$. Compared to DG, its accuracy is one order lower, but to its advantage, the CFL limit is roughly twice as large. It is also simpler than the schemes using Lobatto points in the literature (e.g., Hesthaven 1997, Kopriva 1994, and Huang, Wang, and Liu 2005).

The next result, obtained via Fourier (Von Neumann) analysis, provides a criterion for super-accuracy. If the correction function g is perpendicular to $\boldsymbol{P}_m$ where $0 \le m \le K-2$, then the scheme is accurate to order $K+m+1$. In particular, the DG scheme corresponds to the choice of Radau polynomial for g and, since this polynomial is orthogonal to $\boldsymbol{P}_{K-2}$, the scheme is accurate to order $2K-1$, a fact consistent with its super-accuracy.

Concerning stability, Fourier analysis for values of K up to 10 leads to the following observation: schemes that are accurate to order $2K-1$ and $2K-2$ are stable; schemes accurate to order $2K-3$ or lower usually encounter instabilities which, for well constructed schemes, are mild. For optimal or 'best of class' schemes, we have the following trade-off: the higher the super accuracy, the smaller is the CFL limit. Another observation is that a steeper correction function tends to result in a scheme with a smaller CFL limit.

We then show that for advection, schemes via flux reconstruction can be cast in the DG framework. The difference is the space of test functions. For DG, this space is $\boldsymbol{P}_{K-1}$, which is of dimension K. For the scheme defined by the correction function g, this space is also of dimension K, but consists of polynomials ϕ of degree K such that $d\phi/dx$ is perpendicular to g. If g is the Radau polynomial, the scheme reduces to DG, and this space reduces to $\boldsymbol{P}_{K-1}$.

Moving on to the case of two dimensions, the new flux reconstruction approach can be extended to quadrilateral meshes. Fourier analysis on a rectangular mesh reveals that for schemes via flux reconstruction, the order of accuracy of the scheme is identical to that of the one-dimensional case. The CFL limit, however, is reduced by a factor of $1/\sqrt{2}$, and this reduction takes place due to the case of flow along the diagonal direction. While this reduction in CFL limit for the piecewise constant scheme is well known, the result for the high-order case is new. Finally, the extension to the case of a triangular mesh remains a topic of future research since the concept of tensor product no longer applies.

In the hope of attracting the interest of researchers who are not familiar with these types of high-order schemes, this paper is written in an essentially self-contained manner. It is organized as follows. Section II deals with conservation laws, upwind fluxes, and various flux values to the left and right of each interface. Section III introduces the correction function g. Section IV describes various choices of correction functions. Fourier analysis for a general correction function is provided in Section V. Stability and accuracy analyses of schemes corresponding to various correction functions are given in Section VI. Section VII includes two proofs: (a) the flux reconstruction scheme with g defined by the Radau polynomial is identical to the DG scheme and (b) for advection, flux reconstruction schemes with arbitrary g can be cast in the DG framework. Section VIII discusses 2D extensions. Section IX contains numerical tests. Conclusions and discussions can be found in Section X. Finally, reviews of the Legendre, Radau, and Lobatto polynomials, and corresponding quadrature points, as well as derivations of new polynomials and properties needed are provided in the Appendix.

II. Conservation Laws and Upwind Fluxes

Consider the conservation law

$$u_t + f_x = 0 \tag{2.1}$$

with initial condition $u(x,0) = u_{\text{init}}(x)$ where $u_{\text{init}}(x)$ is periodic or of compact support so that boundary conditions are trivial. The flux f is assumed to depend on u. Denoting the signal speed df/du by $a(u)$, the above can be cast in nonconservation form:

$$u_t + a u_x = 0 . \tag{2.2}$$

Let the domain of calculation be divided into (possibly nonuniform) cells or elements E_j, $j = ... -1, 0, 1, 2, ...$ On each cell E_j, let the solution be approximated by *K* pieces of data $u_{j,k}$, $k = 1, ..., K$ at locations $x_{j,k}$, which are called solution points. The *K* solution points are typically the Gauss or Lobatto points, but equidistant points can also be employed. In fact, the Fourier stability and accuracy results below are independent of the type of points chosen. We assume that the same type of points is employed for all cells. Note that if the Lobatto points are selected, then since they include the two cell interfaces, each cell interface $x_{j+1/2}$ corresponds to two values of *u*, namely, $u_{j,K}$ and $u_{j+1,1}$, and these left and right interface values are readily available for the upwind flux calculation.

At each solution point, the solution $u_{j,k}(t)$ depends on t. For simplicity, $u_{j,k}(t^n)$ is abbreviated to $u_{j,k}$. At time level *n*, suppose the data $u_{j,k}$ are known for all *j* and *k*. We wish to calculate $d u_{j,k}(t)/dt$ at time $t = t^n$, which is abbreviated to $d u_{j,k}/dt$. In other words, we wish to evaluate f_x at the solution points $x_{j,k}$ in terms of the data. Then, we march in time by, say, a Runge-Kutta method.

As is standard in finite element methods (Hughes 2000), instead of dealing with the cell E_j (global element), it is often more convenient to deal with the interval $I = [-1,1]$ (local element). Denote the center of E_j by x_j and its width, h_j. With ξ varying on I and x, on E_j, the linear function that maps I onto E_j and its inverse are

$$x(\xi) = x_j + \xi h_j / 2 \quad \text{and} \quad \xi(x) = 2(x - x_j)/h_j . \tag{2.3a, b}$$

The solution points on I (in the local description) are denoted by ξ_k, $k = 1, ..., K$. They relate to the solution points on E_j (in the global description) by

$$x_{j,k} = x(\xi_k) = x_j + \xi_k h_j / 2 . \tag{2.4}$$

A function $r_j(x)$ on E_j in the global description results in a function in the local description denoted by $r_j(\xi)$ for simplicity of notation:

$$r_j(\xi) = r_j(x(\xi)) = r_j(x) . \tag{2.5}$$

To avoid confusion, the function r_j is usually referred to as $r_j(x)$ or $r_j(\xi)$. The derivatives in the global and local descriptions are related by the chain rule,

$$\frac{dr_j(x)}{dx} = \frac{2}{h_j}\frac{dr_j(\xi)}{d\xi} . \tag{2.6}$$

Returning to solving the conservation law (2.1), the first task is to approximate *u* in the cell E_j by a polynomial of degree $K-1$, which interpolates the *K* pieces of data $u_{j,k}$, $k = 1, ..., K$. To this end, for each *k*, let $\phi_{j,k}$ be the Lagrange polynomial on E_j of degree $K-1$ that takes value 1 at $x_{j,k}$ and 0 at all other solution points. These $\phi_{j,k}$ are called basis (or cardinal or shape) functions. In global and local descriptions respectively, they are

$$\phi_{j,k}(x) = \prod_{l=1,\ l\neq k}^{K} \frac{x - x_{j,l}}{x_{j,k} - x_{j,l}} \quad \text{and} \quad \phi_k(\xi) = \prod_{l=1,\ l\neq k}^{K} \frac{\xi - \xi_l}{\xi_k - \xi_l} . \tag{2.7a, b}$$

On E_j, let $u_j(x)$ be the polynomial of degree $K-1$ interpolating $u_{j,k}$, $k = 1, ..., K$; $u_j(x)$ is called a *solution polynomial*. In global and local coordinates, respectively,

$$u_j(x) = \sum_{k=1}^{K} u_{j,k}\,\phi_{j,k}(x) \quad \text{and} \quad u_j(\xi) = \sum_{k=1}^{K} u_{j,k}\,\phi_k(\xi) . \tag{2.8a, b}$$

Basis functions for the case $K = 4$ using Lobatto points and the solution polynomials are shown in Fig. 2.1.

Next, we define the discontinuous flux function. From the data $u_{j,k}$ at the solution points, we can easily evaluate the corresponding fluxes:

$$f_{j,k} = f(u_{j,k}) . \tag{2.9}$$

Similar to (2.8), we define the flux polynomial $f_j(x)$ to be the polynomial of degree $K-1$ interpolating the values $f_{j,k}$ at the solution points, $k = 1, ..., K$. Again in global and local coordinates, respectively,

$$f_j(x) = \sum_{k=1}^{K} f_{j,k}\,\phi_{j,k}(x) \qquad \text{and} \qquad f_j(\xi) = \sum_{k=1}^{K} f_{j,k}\,\phi_k(\xi)\,. \tag{2.10a, b}$$

Note that both the flux and solution polynomials $f_j(x)$ and $u_j(x)$ are of degree $K-1$. The flux polynomials $f_j(x)$, $j = \ldots -1, 0, 1, 2, \ldots$ form a function, which is generally *discontinuous* across cell interfaces and is called the *discontinuous flux function*.

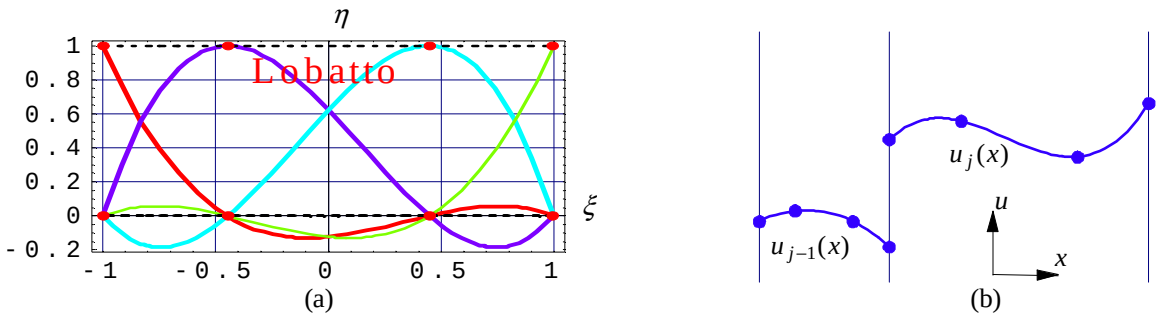


Figure 2.1. (a) Cubic basis functions. *Here, the solution points are the Lobatto points on* $I = [-1,1]$ *for* $K = 4$. *Note that at each interior Lobatto point, the corresponding basis function reaches a local maximum.* **(b) Solution polynomials.** *Cubics* $u_{j-1}(x)$ *and* $u_j(x)$.

If we ignore the jumps and calculate and employ the derivative of the discontinuous flux function, we obtain erroneous results. In other words, for the conservation law (2.1), if we employ $(f_j)_x$ to evaluate f_x, we get poor solutions because such a derivative does not include the interaction of the data between adjacent cells. With such a derivative, the solution never feels the effect of the boundary conditions. In fact, it can be worse as shown in the following example. If $K = 1$, i.e., the case of a constant function in each cell, then by ignoring the jumps, we have $(f_x)_{j,1} = 0$ for all cells; as a result, the solution does not change with time, which is incorrect. To account for the interaction, i.e. the jump, we will construct a *continuous* flux function, which approximates the discontinuous function in some sense, and then calculate its derivative. The continuous flux function will be obtained by adding a correction to the discontinuous one. As a result, we still need the derivative of the discontinuous function, which is given below. At each solution point ξ_k of the local coordinate, $1 \le k \le K$,

$$(f_\xi)_{j,k} = \sum_{l=1}^{K} d_{kl}\, f_{j,l}\,. \tag{2.11}$$

That is, $(f_\xi)_{j,k}$ is given by the dot product of the $(k-\text{th row})$ vector $\{d_{kl}\}_{l=1}^{K}$ with the (column) vector $\{f_{j,l}\}_{l=1}^{K}$. The coefficients d_{kl} form a $K \times K$ matrix denoted by $\boldsymbol{D}$ and is called the derivative matrix (for the derivative operator on I). This matrix can easily be calculated via a software package such as *Mathematica* or *Maple* using (2.10). By (2.6) or the chain rule, for the discontinuous flux function, in the global coordinate,

$$(f_x)_{j,k} = \frac{2}{h_j} \sum_{l=1}^{K} d_{kl}\, f_{j,l}\,. \tag{2.12}$$

For each j, denote the flux and the flux derivative vectors, which are column vectors of K entries, respectively by

$$\boldsymbol{f}_j = \{f_{j,k}\}_{k=1}^{K} \qquad \text{and} \qquad (\boldsymbol{f}_x)_j = \{(f_x)_{j,k}\}_{k=1}^{K}\,. \tag{2.13a,b}$$

Then, the vector of derivatives of the discontinuous flux function at the solution points is given by

$$(\boldsymbol{f}_x)_j = \frac{2}{h_j}\, \boldsymbol{D}\, \boldsymbol{f}_j\,. \tag{2.14}$$

Another way to evaluate $\{(f_x)_{j,k}\}$ is by the chain rule: with $\{(u_x)_{j,k}\}_{k=1}^{K}$ given by a formula similar to the above,

$$(f_x)_{j,k} = (f_u)_{j,k}(u_x)_{j,k} = a(u_{j,k})(u_x)_{j,k}\,. \tag{2.15}$$

The continuous flux function to be defined is required to take the upwind flux values at the interfaces. In addition to the upwind flux values, we also need various left and right interface values discussed below. At the interface $x_{j+1/2}$, let the subscripts L and R denote the values just to its left and right, respectively. That is, let u_L and u_R be the values at $x_{j+1/2}$ taken on by the solution polynomials $u_j(x)$ and $u_{j+1}(x)$ respectively,

$$u_L = u_{j+1/2,L} = u_j(x_{j+1/2}) \qquad \text{and} \qquad u_R = u_{j+1/2,R} = u_{j+1}(x_{j+1/2}). \tag{2.16a, b}$$

Similarly, let f_L and f_R be the values at $x_{j+1/2}$ taken on by the flux polynomials $f_j(x)$ and $f_{j+1}(x)$ respectively,

$$f_L = f_{j+1/2,L} = f_j(x_{j+1/2}) \qquad \text{and} \qquad f_R = f_{j+1/2,R} = f_{j+1}(x_{j+1/2}). \tag{2.17a, b}$$

In the local coordinate ξ on $I = [-1,1]$, we have

$$f_{j+1/2,L} = f_j(1) \qquad \text{and} \qquad f_{j+1/2,R} = f_{j+1}(-1). \tag{2.17c, d}$$

In general, $u_L \neq u_R$, and $f_L \neq f_R$. Since u_L is defined by $u_j(x)$, f_L by $f_j(x)$, u_R by $u_{j+1}(x)$, and f_R by $f_{j+1}(x)$, again, in general,

$$f_L \neq f(u_L) \qquad \text{and} \qquad f_R \neq f(u_R). \tag{2.18a, b}$$

We must exercise caution in the choice of using $f(u_L)$ or f_L. At each interface, as will be shown, we use $f(u_L)$ and $f(u_R)$ for the definition of the upwind flux. For the jump in flux value employed with the correction function, however, we use $f_{\text{upw}} - f_L$ and $f_{\text{upw}} - f_R$ as will be shown in (3.8) to assure that the scheme is conservative.

We can now derive one of the most popular upwind fluxes, namely, Roe's flux (Roe 1986). At the interface $x_{j+1/2}$, with u_L and u_R given by (2.16), let $\tilde{u}$ be the value that satisfies the mean value theorem:

$$a(\tilde{u}) = \begin{cases} (f(u_R) - f(u_L))/(u_R - u_L) & \text{if} \quad u_L \neq u_R, \\ a(u_L) = a(u_R) & \text{otherwise.} \end{cases} \tag{2.19}$$

(For the Burgers equation, $f(u) = u^2/2$; consequently, $\tilde{u} = (u_L + u_R)/2$.) The upwind flux is determined by the wind direction, i.e., the sign of $a(\tilde{u})$:

$$f_{\text{upw}} = f_{j+1/2,\text{upw}} = \begin{cases} f(u_L) & \text{if} \quad a(\tilde{u}) \geq 0, \\ f(u_R) & \text{otherwise.} \end{cases} \tag{2.20}$$

In the above expression, we split the flux based on the sign of $a(\tilde{u})$. If $a(\tilde{u}) = 0$, then by (2.19), $f(u_L) = f(u_R)$. As a result, it does not matter which of $f(u_L)$ and $f(u_R)$ is selected. By using (2.19) and unifying the cases of $a(\tilde{u}) \geq 0$ and $a(\tilde{u}) < 0$, the above upwind flux can be written with no conditional statement:

$$f_{\text{upw}} = \tfrac{1}{2}[f(u_L) + f(u_R)] - \tfrac{1}{2}|a(\tilde{u})|(u_R - u_L). \tag{2.21}$$

This expression is called flux difference splitting.

Note that if we split the flux, then when $a(\tilde{u}) = 0$, it is still well defined. However, if we split the variable u, then when $a(\tilde{u}) = 0$, it is not clear which of u_L or u_R should be selected. Therefore, we do not split u.

For the Euler equations, $\tilde{u}$ in (2.19) turns into Roe's tilde state, and the upwind flux via (2.20) or (2.21) becomes Roe's flux. Note in passing that for Roe's scheme, the condition $\Delta \boldsymbol{F} = \tilde{\boldsymbol{A}} \Delta \boldsymbol{U}$ (similar to (2.19)) between the flux and conservative variables is well known; however, the condition $\Delta \boldsymbol{U} = \tilde{\boldsymbol{M}} \Delta \boldsymbol{V}$ between the conservative and primitive variables, which simplifies the derivation, is not (see Huynh 1995). Also note that the upwind flux given by (2.21) generally requires an entropy correction, which is beyond the scope of this paper.

III. Flux Correction Functions

The upwind flux can be employed in (at least) two different approaches. The first approach deals with the integral form. Here, after multiplying the differential conservation law with a test function, integrating over a cell, and applying integration by parts, the upwind fluxes are employed to evaluate the boundary integral. This approach results in the discontinuous Galerkin or DG methods (see Cockburn, Karniadakis, and Shu 2000 and the references therein) and the discontinuous spectral element methods (Kopriva, Woodruff, and Hussaini 2000). The second approach solves the differential form of the conservation law. Here, to account for the data interaction between adjacent cells, we construct a *continuous* function denoted by $F(x)$ for the flux, which takes the upwind flux values

at the cell interfaces. In each cell j, $F(x)$ is a polynomial denoted by $F_j(x)$, which approximates the discontinuous flux function $f_j(x)$ defined in (2.10). The staggered-grid scheme of Kopriva and Kolias (1996) and spectral difference scheme of Liu, Vinokur, and Wang (2003, 2005) employ this approach and reconstruct the continuous flux function $F(x)$ by requiring it to interpolate the fluxes at the $K-1$ interior Chebyshev-Lobatto points.

Note that the concept of 'recovery' presented by Van Leer and Nomura (2005) and Van Leer, Lo, and Van Raalte (2007) for the diffusion problem and the concept of 'reconstruction' discussed in this paper are similar in that both approaches seek to construct a function that represents the data better than the discontinuous piecewise polynomial function. The key difference, however, is that the 'recovery' spans across adjacent cells whereas the 'reconstruction' stays within a cell (but includes the two cell interfaces).

We now introduce a new flux reconstruction approach, which is considerably more general than that of the staggered grid scheme. The approach leads to several new schemes as well as simplified versions of the DG and staggered-grid methods. As a result of the new approach, the DG scheme can be formulated using the differential form, and such a formulation involves no quadratures. For efforts to avoid quadratures, see (Atkins and Shu, 1998) and (Dumbser and Munz, 2007).

To describe the approach, first, as mentioned above, the continuous flux function $F(x) = F_j(x)$ is required to take the upwind flux values at the two cell interfaces:

$$F_j(x_{j-1/2}) = f_{j-1/2,\text{upw}} \qquad \text{and} \qquad F_j(x_{j+1/2}) = f_{j+1/2,\text{upw}}\,. \tag{3.1a, b}$$

Next, what degree should the polynomial $F_j(x)$ be? To answer this question, in the cell j, we allow the solution polynomial $u_j(x)$ to depend on t, i.e., for each fixed t, $u_j(x,t)$ is a polynomial of degree $K-1$. The conservation law (2.1) can then be approximated by

$$\frac{\partial}{\partial t}u_j(x,t) + (F_j(x))_x = 0\,. \tag{3.2}$$

After one (Euler explicit) time step of size τ (one stage of the explicit Runge-Kutta time stepping), denoting the solution polynomial at the new time by $u_j^{\text{new}}(x)$, the above implies

$$u_j^{\text{new}}(x) = u_j(x) - \tau(F_j(x))_x\,. \tag{3.3}$$

Here, the right hand side is evaluated at time t^n, and $u_j(x)$ is of degree $K-1$. For the solution $u_j^{\text{new}}(x)$ to also be of degree $K-1$, since τ is a constant, the function $(F_j(x))_x$ must be a polynomial of degree $K-1$ in x. Therefore, $F_j(x)$ is required to be of degree K. It is also required to approximate $f_j(x)$, which is of degree $K-1$.

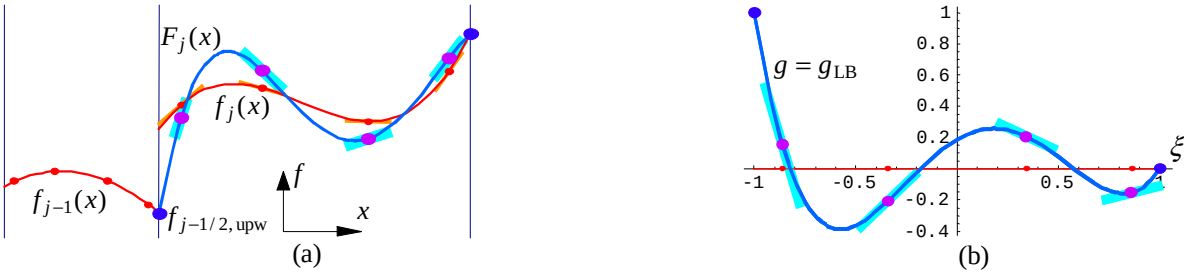


Figure 3.1. (a) Discontinuous and continuous flux functions. *Here, the conservation law is the advection equation $u_t + u_x = 0$. The solution points are the Gauss points with $K = 4$. The curve interpolating flux values at the solution points (small dots) is the discontinuous flux function $f_j(x)$, which, in this case, is the same as the solution polynomial. It is of degree $K-1$. At the interface, the upwind flux is the flux from the left cell. The (blue) curve connecting the big dots is the continuous flux function $F_j(x)$, which is a polynomial of degree K. At each solution point, f_x of the conservation law is approximated by the derivative of $F_j(x)$ as shown by the thick (light blue) line segments. Each of these slopes can be obtained by adding a correction term to the slope of the discontinuous flux function.* **(b) A correction function for the left boundary**. *Here, g is the right Radau polynomial. The thick line segments correspond to the values of the derivative of g at the solution points. Note the alternating signs of the slopes.*

Note that in (3.3), both sides are polynomials of degree $K-1$. These polynomials can be described by using K solution points of any type. The accuracy and stability results via Fourier analysis, as will be shown in Section 6, are independent of the type of solution points chosen. In fact, similar to the DG scheme, instead of using the solution points, we can use the Legendre polynomials as basis functions as well. The derivative of the correction function g must then be expressed in terms of the Legendre polynomials. For the 1D case, the Legendre polynomials work well; for the 2D case, however, the flux reconstruction schemes using the Legendre polynomials are not as simple as those using solution points.

As discussed above, the continuous flux function $F(x)$ is required to take the upwind flux values at cell interfaces. In each cell E_j, it is a polynomial of degree K denoted by $F_j(x)$, and it approximates the discontinuous flux function $f_j(x)$. Instead of defining $F(x)$, we define $F_j(x)-f_j(x)$, which approximates the function 0. Switching to the local coordinate ξ on $I=[-1,1]$, at the two interfaces $\xi=\pm 1$, the function $F_j(\xi)-f_j(\xi)$ takes the following values which, loosely put, are the jumps:

$$F_j(-1)-f_j(-1)=f_{j-1/2,\mathrm{upw}}-f_j(-1) \qquad \text{and} \qquad F_j(1)-f_j(1)=f_{j+1/2,\mathrm{upw}}-f_j(1). \tag{3.4a, b}$$

Therefore, $F_j(\xi)-f_j(\xi)$ is a polynomial of degree K with prescribed left and right interface values.

We now separate the prescription of the jump at the left interface from that of the right. Again, using the local coordinate ξ on I, for the left interface, let g_{LB} be the *correction function* (LB for 'left boundary') defined by

$$g_{\mathrm{LB}}(-1)=1, \quad g_{\mathrm{LB}}(1)=0, \tag{3.5a, b}$$

and g_{LB} is a polynomial of degree K approximating the constant function 0 in some sense. Similarly, let g_{RB} be the correction function for the right boundary:

$$g_{\mathrm{RB}}(-1)=0, \quad g_{\mathrm{RB}}(1)=1, \tag{3.6a, b}$$

and g_{RB} is also of degree K and approximating 0 in some sense. Consider the left interface $x_{j-1/2}$, i.e., $\xi=-1$. The polynomial

$$f_j(\xi)+[f_{j-1/2,\mathrm{upw}}-f_j(-1)]\,g_{\mathrm{LB}}(\xi) \tag{3.7}$$

provides a correction at the left interface for $f_j(\xi)$ by changing the flux value at this interface from $f_j(-1)$ to $f_{j-1/2,\mathrm{upw}}$, while leaving the value at the right interface unchanged, namely, $f_j(1)$. Next, the polynomial

$$F_j(\xi) = f_j(\xi)+[f_{j-1/2,\mathrm{upw}}-f_j(-1)]\,g_{\mathrm{LB}}(\xi)+[f_{j+1/2,\mathrm{upw}}-f_j(1)]\,g_{\mathrm{RB}}(\xi) \tag{3.8}$$

provides corrections to both interfaces: using (3.5) and (3.6), one can easily verify that $F_j(-1)=f_{j-1/2,\mathrm{upw}}$ and $F_j(1)=f_{j+1/2,\mathrm{upw}}$. Thus, the above $F_j(\xi)$ is of degree K, takes the upwind flux values at the two interfaces, and approximates $f_j(\xi)$ in the same sense that g_{LB} and g_{RB} approximate the zero function.

Finally, the derivative of $F_j(\xi)$ at the solution point ξ_k is

$$(F_\xi)_{j,k} = (f_\xi)_{j,k}+[f_{j-1/2,\mathrm{upw}}-f_j(-1)]\,g_{\mathrm{LB}}{}'(\xi_k)+[f_{j+1/2,\mathrm{upw}}-f_j(1)]\,g_{\mathrm{RB}}{}'(\xi_k). \tag{3.9}$$

In the global coordinate, at the solution point $x_{j,k}$,

$$(F_x)_{j,k} = \frac{2}{h_j}(F_\xi)_{j,k}. \tag{3.10}$$

The quantities $(F_x)_{j,k}$ given by (3.9) and (3.10) are employed to approximate f_x at the solution point $x_{j,k}$ for the conservation law. Finally,

$$\frac{du_{j,k}}{dt}=\left[\frac{du_{j,k}(t)}{dt}\right]_{t=t^n}=-(F_x)_{j,k}. \tag{3.11}$$

The solution $u_{j,k}$ can then be updated via an explicit Runge-Kutta method.

What is crucial in (3.9) is the following. At each solution point, the derivative $(F_\xi)_{j,k}$ of the continuous flux function is obtained by correcting the derivative $(f_\xi)_{j,k}$ of the discontinuous flux function. The correction amount

is straightforward once the derivatives $g_{\mathrm{LB}}'(\xi_k)$ and $g_{\mathrm{RB}}'(\xi_k)$ are known for $1 \le k \le K$. These derivatives, in turn, can easily be derived once g_{LB} and g_{RB} are defined on I.

We summarize the algorithm for the new flux reconstruction schemes below.

Algorithm. At time level n, suppose the data $u_{j,k}$ are known for all j and k.

(1) At each interface $j+1/2$, if the left and right values $u_{j+1/2,L}$ and $u_{j+1/2,R}$ are not available, calculate them; then obtain and store the upwind flux at all interfaces via (2.20) or (2.21).

(2) In the cell j, for $k = 1,...,K$, evaluate the fluxes $f_{j,k} = f(u_{j,k})$ at the K solution points. Next, using these fluxes, obtain the derivatives of the discontinuous flux function in the local coordinate, i.e., $(f_\xi)_{j,k}$, by (2.11).

(3) At the two interfaces of cell j, calculate the jumps $f_{j-1/2,\mathrm{upw}} - f_j(-1)$ and $f_{j+1/2,\mathrm{upw}} - f_j(1)$. For $k = 1,...,K$, evaluate the derivatives of the continuous flux function in the local coordinate namely, $(F_\xi)_{j,k}$, by (3.9). Then obtain $(F_x)_{j,k}$ by (3.10).

(4) Using the corrected slope $(F_x)_{j,k}$, calculate $du_{j,k}/dt$ by (3.11) and march in time by a Runge-Kutta method. This completes the algorithm.

Next, we show that the resulting scheme is conservative, that is, we wish to show

$$\left[\frac{\partial}{\partial t}\int_{E_j} u_j(x,t)\,dx\right]_{t=t^n} = f_{j-1/2,\mathrm{upw}} - f_{j+1/2,\mathrm{upw}}\,. \tag{3.12}$$

To this end, consider the quadrature of degree of precision at least $K-1$ using the K solution points, i.e., it is exact for polynomial of degree $K-1$ or less. Denote the weight at ξ_k by w_k. Then, since $u_j(\xi)$ is of degree $K-1$,

$$\int_{-1}^{1} u_j(\xi)\,d\xi = \sum_{k=1}^{K} w_k\, u_{j,k}\,. \tag{3.13}$$

Note that the above is an equation, not an approximation. For the global coordinate,

$$\int_{E_j} u_j(x)\,dx = \frac{h_j}{2}\sum_{k=1}^{K} w_k\, u_{j,k}\,. \tag{3.14}$$

The left hand side of (3.12) can then be written as

$$\left[\frac{\partial}{\partial t}\int_{E_j} u_j(x,t)\,dx\right]_{t=t^n} = \frac{h_j}{2}\frac{\partial}{\partial t}\sum_{k=1}^{K} w_k\, u_{j,k} = \frac{h_j}{2}\sum_{k=1}^{K} w_k \frac{du_{j,k}}{dt} = -\frac{h_j}{2}\sum_{k=1}^{K} w_k\,(F_x)_{j,k}\,. \tag{3.15}$$

Here, the last equation is a consequence of (3.11). Since $(F_j)_x$ is a polynomial of degree $K-1$, using the quadrature formula (3.14),

$$\int_{E_j} (F_j)_x(x)\,dx = \frac{h_j}{2}\sum_{k=1}^{K} w_k\,(F_x)_{j,k}\,. \tag{3.16}$$

By the fundamental theorem of Calculus,

$$\int_{E_j} (F_j(x))_x\,dx = F_j(x_{j+1/2}) - F_j(x_{j-1/2}) = f_{j+1/2,\mathrm{upw}} - f_{j-1/2,\mathrm{upw}}\,. \tag{3.17}$$

The above three equations then imply (3.12). This completes the proof.

From here on, we focus only on g_{LB} since g_{RB} follows by symmetry. For simplicity of notation, when there is no confusion, we drop the subscript and denote g_{LB} by g. Recall that a polynomial of degree K can have no more than K zeros. The right interface $\xi = 1$ is a zero; as a result, g is completely defined if we specify $K-1$ additional zeros. For examples, for the case $K = 2$ (piecewise linear or linear element), g is a quadratic defined by (3.5) and one additional zero; for $K = 4$ (cubic element), g is of degree four and defined by again (3.5) and three additional zeros. See Fig. 3.1(b).

IV. Choices of Correction Functions

This section employs the Legendre, Radau, and Lobatto polynomials and some of their main properties. Since these polynomials and properties play a key role, they are reviewed in the Appendix. We also use several new polynomials and properties, and these are derived in the Appendix as well.

Under the above approach of flux reconstruction, the problem of constructing high-order schemes reduces to defining a correction function g on $I=[-1,1]$, which is a polynomial of degree K determined by $g(-1)=1$, $g(1)=0$, and $K-1$ additional conditions. These conditions are prescribed so that g approximates the function 0 in some sense. But what criteria should we employ? The final criteria are the stability and accuracy of the resulting scheme discussed in Section 6. Since there is generally a trade-off between accuracy and stability, and the property of superaccuracy does not hold for the general case of the Euler or Navier-Stokes equations, an optimal scheme has not been determined by this author. In fact, such a scheme is likely to be problem dependent. In this section, therefore, we present various choices for g via approximation theory as well as some guidance from Fourier analysis. To distinguish the several correction functions, they are assigned a subscript. We employ a subscript of letters if the correction function leads to an existing scheme e.g., g_{DG} and g_{SG} (staggered-grid) or if it involves well-known points, e.g., g_{Lo} (Lobatto), g_{Ga} (Gauss). We also use integer subscripts, e.g., $g_1,...,g_K$. For a more precise notation, we need two subscripts, e.g., $g_{\mathrm{DG},K}$ or $g_{2,K}$, but the subscript K for the number of solution points is understood. Unless otherwise stated, we use the notations $\xi_1,...,\xi_K$ for the solution points, and $\zeta_1,...,\zeta_{K-1}$, and $\zeta_K=1$ for the zeros of the correction function.

The first choice for g, denoted by g_{DG} or g_1 (the right boundary is a zero of multiplicity one), is derived by the following argument. Since g is of degree K and approximates the function 0, using the criterion of least squares, we could require that the projection of g onto $\boldsymbol{P}_{K-1}$ is 0, where $\boldsymbol{P}_{K-1}$ is the space of polynomials of degree $K-1$ or less. This requirement would imply that $g=\text{constant}\times P_K$, where P_K is the Legendre polynomial of degree K. Since $P_K(-1)=(-1)^K$ and $P_K(1)=1$, the conditions $g(-1)=1$ and $g(1)=0$ cannot be met simultaneously no matter which constant we choose. Therefore, the highest accuracy possible (via least squares) is the requirement that the projection of g onto $\boldsymbol{P}_{K-2}$ is 0. This requirement is met by $R_{R,K}$, the right Radau polynomial of degree K shown in Fig. 4.1(a). The zeros of the right Radau polynomial are the right Radau points. Note that the correction function for the *left* boundary is the *right* Radau polynomial, which vanishes at $\xi=1$.

The following remarks are in order. The requirement that g is perpendicular to $\boldsymbol{P}_{K-2}$ provides $K-1$ conditions in determining g; the other two conditions are $g(-1)=1$ and $g(1)=0$. As will be shown in Section 7, the resulting scheme turns out to be identical to the DG method; for this reason, this correction function is denoted by g_{DG}. Later, we will show via Fourier analysis that the scheme is accurate to order $2K-1$, a fact consistent with the super-accuracy of the DG scheme. The function g_{DG} (a more precise notation is $g_{\mathrm{DG},K}$) can be expressed in terms of the Legendre polynomial as

$$g_{\mathrm{DG}}=g_{\mathrm{DG},K}=R_{R,K}=\frac{(-1)^K}{2}(P_K-P_{K-1}). \tag{4.1}$$

Since the Legendre polynomial P_K is perpendicular to $\boldsymbol{P}_{K-1}$, the above expression for g_{DG} shows that it is perpendicular to $\boldsymbol{P}_{K-2}$. While the slopes at the solution points must generally be evaluated numerically, the slopes at the two boundaries are given by the following formula:

$$g_{\mathrm{DG}}'(-1)=R_{R,K}'(-1)=-\frac{K^2}{2} \qquad \text{and} \qquad g_{\mathrm{DG}}'(-1)=R_{R,K}'(1)=(-1)^{K-1}\frac{K}{2}. \tag{4.2}$$

The next correction function, denoted by g_{SG}, results in a simplified version of Kopriva and Kolias' staggered-grid scheme (1996). It is derived as follows. Consider the $K+1$ Chebyshev-Lobatto points on I defined by, for $k=0,...,K$,

$$\zeta_k=-\cos(k\pi/K). \tag{4.3}$$

The function g_{SG} is required to take value 1 at $\zeta_0=-1$ and 0 at all other ζ_k, $k=1,...,K$. See Fig. 4.1(b). The resulting scheme is more economical than the original staggered-grid version since we only need one grid, namely,

the solution points; the Chebyshev-Lobatto (or flux) points are used only in the derivation of g and not in the coding of the method. As will be shown in Section 6, compared to DG, the staggered-grid scheme is less accurate, but to its advantage, the CFL limit is more than 3/2 times as large. Although the scheme had the disadvantage of being mildly unstable, this problem will be remedied by one of the new methods that share the symmetry property discussed below.

In general, if the $K-1$ interior zeros of the correction function, namely ζ_k, $k=1,...,K-1$, are *symmetric* about 0, then both g_{LB} and g_{RB} have the same interior zeros; therefore, $F_j(\zeta_k)=f_j(\zeta_k)$ where F_j is given by (3.8). As a result, the continuous flux function $F_j(x)$ can be defined by the two interface upwind fluxes and the values $f_j(\zeta_k)$, $k=1,...,K-1$ as in (Kopriva and Kolias 1996). The formulation here, again, is more economical since these $f_j(\zeta_k)$ are not needed.

As shown in Fig. 4.1(b), loosely put, g_{SG} is less steep than g_{DG}. Generally, a steeper correction function results in a scheme with a smaller CFL limit.

Another difference between the current version and the original staggered-grid scheme is the following. For the staggered-grid approach, one first interpolates the values $u_{j,k}$ at the solution points to obtain the u values at the interior flux points, and then, calculates the corresponding flux values at these flux points. For the current approach, these flux values, though never explicitly evaluated, correspond to first calculating the fluxes at the solution points, then interpolating these fluxes via the flux polynomial (discontinuous flux function) $f_j(x)$ to the flux points.

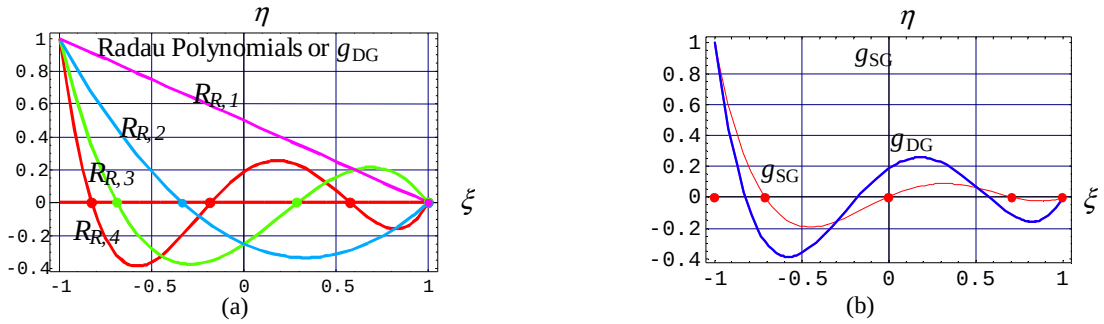


Figure 4.1. Correction functions. **(a)** *The first choice for* g *denoted by* g_{DG} *or, more precisely,* $g_{\mathrm{DG},K}$ *is the right Radau polynomial* $R_{R,K}$*; here, we plot* $R_{R,K}$ *for* $K=1,...,4$*. The resulting scheme yields a simplified version of the DG scheme.* **(b)** *Correction function* g_{SG} *for the case* $K=4$*. The resulting scheme is a simplified version of Kopriva and Kolias' staggered-grid method. Note that the interior zeros of* g_{SG} *(the dots excluding the ones on the boundaries) are symmetric about 0. Also shown is* g_{DG} *for comparison. Loosely put,* g_{SG} *is less steep than* g_{DG}*. Generally, a steeper correction function results in a scheme with a smaller CFL limit.*

We now discuss correction functions that result in new schemes.

The next g, denoted by g_2 or $g_{\mathrm{Lump,\,Lo}}$ (for 'lumping for Lobatto points ' explained later), is defined as follows. Recall that g_{DG} is perpendicular to $\boldsymbol{P}_{K-2}$. If we relax this condition to the requirement that g is perpendicular to $\boldsymbol{P}_{K-3}$, then we have one degree of freedom to experiment with. Since a steeper correction function tends to result in a scheme with a smaller CFL limit, we wish to make g less steep. On the other hand, since g approximates the function 0 on I, it seems reasonable to require that all zeros of g lie on I. To make g less steep, therefore, the extra condition is obtained by pushing one of the zeros to the right boundary. Thus, g is determined by the conditions that it is perpendicular to $\boldsymbol{P}_{K-3}$, $g(-1)=1$, $g(1)=0$, and $g'(1)=0$. Since $\xi=1$ is a root (zero) of multiplicity two, this correction function is denoted by g_2 and shown in Fig. 4.2(a).

The correction function g_2 has the following remarkable property, which holds true for any K. Among the K Lobatto points (the zeros of the Lobatto polynomials of degree K), g_2' vanishes at $K-1$ of them; the exception is the left boundary point $\xi=-1$ as shown in Fig. 4.2(a). Therefore, if we employ g_2, it is convenient and

economical to select the K Lobatto points as solution points. The jump in flux values at the left interface results in a correction to only $(f_x)_{j,1}$ and not to any $(f_x)_{j,k}$, $k>1$. That is, the correction due to the jump at the left boundary is lumped into that boundary, and the corrections at all other solution points are zero. For this reason we also use the notation $g_{\text{Lump, Lo}}$. Also note that, $g_2'(-1) = -K(K-1)/2$, which equals $-w_1$ where w_1 is the weight at the boundary point of the Lobatto quadrature with K points. This correction function can be expressed as a weighted average of the Radau polynomials $R_{R,k}$:

$$g_2 = \frac{K-1}{2K-1} R_{R,K} + \frac{K}{2K-1} R_{R,K-1}. \tag{4.4}$$

Via Fourier analysis for values K up to 10, it will be shown that g_2 results in a scheme accurate to order $2K-2$ as opposed to order $2K-1$ of the DG scheme. To its advantage, the CFL limit is roughly twice as large as that of DG.

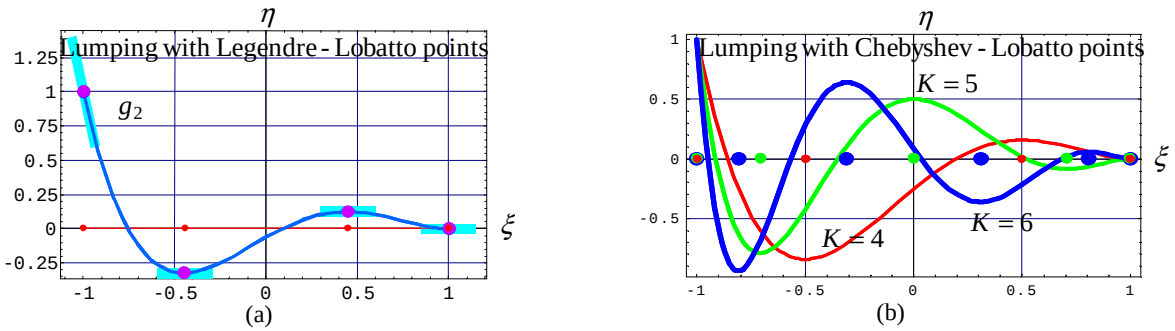


Figure 4.2. Correction functions. **(a)** *If we choose the K Lobatto points as solution points (small dots), then* g_2' *vanishes at all solution points except the left boundary. In other words, the correction is lumped to the left boundary. For this reason,* g_2 *is also denoted by* $g_{\text{Lump, Lo}}$. *As will be shown, the resulting scheme is accurate to order* $2K-2$, *one order lower than that of the DG scheme.* **(b)** *Lumping for the Chebyshev-Lobatto points with* $K = 4, 5,$ *and* 6. *Note the large magnitude of the oscillations and the fact that* g *reaches local extrema at the solution points.*

We can apply the above lumping strategy to any set of solution points $\xi_1, ..., \xi_K$. Indeed, the correction function g, which is a polynomial of degree K, can be defined by requiring that $g(-1) = 1$, $g(1) = 0$, and $g'(\xi_k) = 0$ for $k = 2, ..., K$. That is, we lump the correction to the solution point closest to the left boundary, and $g'(\xi_1)$ can easily be calculated once g is determined. For a certain choice of solution points such as the Legendre-Lobatto points, this requirement yields a stable and accurate scheme. However, in general, the resulting g may be a poor approximation to the function 0, and the scheme may have large errors and/or be unstable.

In addition to the Legendre-Lobatto points, which lead to g_2, we also test the Chebyshev-Lobatto points, i.e., $\xi_k = -\cos(k\pi/K)$, $k = 1, ..., K$. For this choice, the correction function that lumps the correction to the left boundary has somewhat large oscillations: the minimum values for g are less than -0.7. See Fig. 4.2(b).

We now derive a series of correction functions using an argument similar to that in the derivation of g_2. Recall that g is defined by the condition $g(-1) = 1$ and K additional conditions. These K additional conditions are, for $g_1 = g_{\text{DG}}$, the point $\xi = 1$ is a zero of multiplicity one, and g_1 is perpendicular to $\boldsymbol{P}_{K-2}$. For g_2, the point $\xi = 1$ is a zero of multiplicity two, and g_2 is perpendicular to $\boldsymbol{P}_{K-3}$. Following this pattern, for any integer $m \geq 1$, in addition to the condition $g(-1) = 1$, the correction function g_m is defined by the K conditions that $\xi = 1$ is a zero of multiplicity m and g_m is perpendicular to $\boldsymbol{P}_{K-m-1}$. See Fig. 4.3(a). For the advection equation, it will be shown via Fourier analysis that the corresponding scheme is (super) accurate to order $2K-m$.

If $m = K-1$, then g_{K-1} is defined by the following conditions. First, the point $\xi = 1$ is a zero of multiplicity $K-1$. Next, since $K-m-1=0$, we have $\boldsymbol{P}_{K-m-1} = \boldsymbol{P}_0$; since $\boldsymbol{P}_0$ is the subspace (in this case, a line) consisting of all constant functions, the condition that g_{K-1} is perpendicular to $\boldsymbol{P}_0$ means

$$(g_{K-1}, 1) = \int_{-1}^{1} g_{K-1}(\xi)\, d\xi = 0 . \tag{4.5}$$

That is, g_{K-1} is defined by $g_{K-1}(-1) = 1$, $\xi = 1$ is a zero of multiplicity $K-1$, and the above equation.

If $m = K$, then the K additional conditions for defining g_K are that the point $\xi = 1$ is a zero of multiplicity K. For this case,

$$g_K(\xi) = \left(\frac{1-\xi}{2}\right)^K . \tag{4.6}$$

Note that g_K is decreasing on I and $g(\xi) \geq 0$ for all ξ in I.

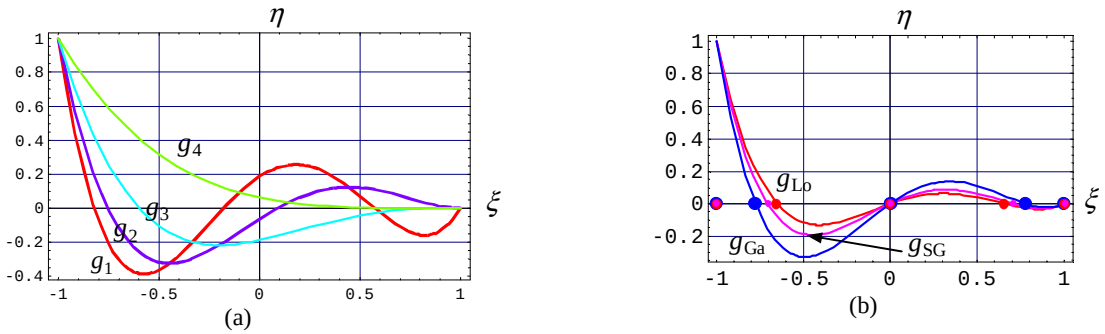


Figure 4.3. More correction functions. **(a)** *Correction functions* g_m, $m = 1, \ldots, K$ *for the case* $K = 4$. *For all tested values of* K *(up to 10),* g_m *results in a scheme accurate to order* $2K-m$. **(b)** *Three correction functions whose interior zeros are symmetric about* $\xi = 0$*; they are* g_{SG} *(using the Chebyshev-Lobatto points),* g_{Lo} *(Legendre-Lobatto), and* g_{Ga} *(Gauss). Here,* g_{Ga} *results in a stable scheme, which is accurate to order* $2K-2$. *The other two results in mildly unstable schemes, which are accurate to order* K.

The following discussion on g_m is in order. First, the slope at the left boundary of g_m are given by, for $1 \leq m \leq K$,

$$g_m'(-1) = -\frac{1}{2} K(K-m+1) . \tag{4.7}$$

In particular, $g_1'(-1) = -K^2/2$ and $g_K'(-1) = -K/2$. The above can also be expressed recursively as follows. With $g_1'(-1) = -K^2/2$, for $1 \leq m \leq K-1$,

$$g_{m+1}'(-1) = g_m'(-1) + (K/2) . \tag{4.8}$$

As an example, for the case $K = 4$ shown in Fig. 4.3(a), we have $g_1'(-1) = -8$, $g_2'(-1) = -6$, $g_3'(-1) = -4$, and $g_4'(-1) = -2$.

For all correction functions tested by this author, with K solution points, if g is of degree K and perpendicular to $\boldsymbol{P}_m$ where $0 \leq m \leq K-2$, then the resulting scheme has order of accuracy $K+m+1$ via Fourier analysis.

The next two correction functions are similar to g_{SG} of the staggered-grid scheme in that their interior zeros, (i.e., excluding $\zeta_K = 1$) are symmetric about $\xi = 0$. For the first of the two, instead of the Chebyshev-Lobatto points, consider the $K+1$ Legendre-Lobatto points $\zeta_0, \ldots, \zeta_K$ where $\zeta_0 = -1$ and $\zeta_K = 1$. Let g_{Lo} (Legendre-Lobatto) be defined by $g(-1) = 1$ and g vanishes at $\zeta_1, \ldots, \zeta_K$. See Fig. 4.3(b). This correction function, similar to g_{SG}, is not perpendicular to any Legendre polynomial P_k, $k = 0, \ldots, K$, i.e., g_{Lo} is not perpendicular to any $\boldsymbol{P}_k$. As will be shown, the resulting scheme is accurate to order K. It is also slightly more unstable than the staggered-grid scheme.

The instability of the schemes corresponding to g_{SG} and g_{Lo} is remedied by the next correction function denoted by g_{Ga} (Gauss) and defined as follows. Consider the $K-1$ Gauss points (the zeros of the Legendre polynomial P_{K-1}). They are completely different from the K Gauss points (the zeros of the Legendre polynomial P_K), which can be chosen as the K solution points. Recall that the quadrature associated with the $K-1$ Gauss points has a degree of precision of $2(K-1)-1=2K-3$. We define g_{Ga} by requiring that its interior zeros are these $K-1$ Gauss points. It follows that g_{Ga} is perpendicular to $\boldsymbol{P}_{K-3}$. Indeed, recall that g_{Ga} is of degree K. Let p be a polynomial of degree $K-3$, then $p\,g_{\mathrm{Ga}}$ is of degree $2K-3$. Since $p\,g_{\mathrm{Ga}}$ vanishes at the $K-1$ Gauss points, due to the degree of precision of the Gauss quadrature, $\int_{-1}^{1} p(\xi)\,g_{\mathrm{Ga}}(\xi)\,d\xi=0$, and the claim that g_{Ga} is perpendicular to $\boldsymbol{P}_{K-3}$ follows.

Since g_{Ga} is perpendicular to $\boldsymbol{P}_{K-3}$, the resulting scheme is (super) accurate to order $2K-2$. To its advantage, the CFL limit, as will be shown, is roughly 3/2 that of DG.

We conclude this section by the following observation. The zeros of g_{Ga} lie closer to the boundaries than those of g_{SG}, which, in turn, are closer to the boundaries than those of g_{Lo} (i.e., $\zeta_{1,\mathrm{Ga}}<\zeta_{1,\mathrm{SG}}<\zeta_{1,\mathrm{LO}}$). Consequently, among these three correction functions, g_{Ga} is, loosely put, the steepest; it will be shown that the resulting scheme has the smallest CFL limit.

V. Fourier (Von Neumann) Analysis

Fourier analysis provides information on both stability and accuracy of schemes. Concerning accuracy however, it is worth keeping in mind that the super-accuracy property of schemes no longer holds for the general case of the Euler or Navier-Stokes equations due to nonlinear errors.

For Fourier analysis, on the domain $(-\infty, \infty)$, the conservation law is the advection equation

$$u_t + u_x = 0\,. \tag{5.1}$$

The initial condition is periodic:

$$u_{\mathrm{init}}(x) = e^{i w x} \tag{5.2}$$

where w is a real number between $-\pi$ and π called a wave number. Low frequency data corresponds to w of small magnitude; high frequency, to w near $\pm\pi$. The exact solution is

$$u_{\mathrm{exact}}(x,t) = e^{i w (x-t)}\,. \tag{5.3}$$

At $x=0$ and $t=0$, the exact time partial derivative is purely imaginary:

$$(u_{\mathrm{exact}})_t(0,0) = -i\,w\,. \tag{5.4}$$

The numerical schemes employ the cells $E_j=[j-1/2,\,j+1/2]$. Denote the solution points in the local coordinate on $I=[-1,1]$ by ξ_k, $k=1,\ldots,K$. The global solution points are

$$x_{j,k} = j + \xi_k/2\,. \tag{5.5}$$

The initial condition (5.2) evaluated at these solution points are

$$u_{j,k} = \exp[\,i\,w(j+\xi_k/2)]\,. \tag{5.6}$$

However, it is not the above expression but the following property of the data, which plays a key role in the calculation of eigenvalues,

$$u_{j-1,k} = e^{-i w} u_{j,k}\,. \tag{5.7}$$

Note that to calculate the eigenvalues, the K solution points are grouped together as a vector, and we must express the solution vector in terms of the data vectors. In this manner, the solution vector for the cell $j-1$ looks the same as that for the cell j except for a shift in the index.

Since the advection speed is 1, the wind blows from the left, and the upwind value at each interface $j+1/2$ is the value from the cell j. As a result, there is no need to carry out the correction for the right boundary, i.e., we only need g_{LB}, which is denoted by g. Next, in the local coordinates ξ on I, set

$$\boldsymbol{\xi} = \begin{pmatrix} \xi_1 \\ \vdots \\ \xi_K \end{pmatrix} \quad \text{and} \quad \boldsymbol{g'} = \begin{pmatrix} g'(\xi_1) \\ \vdots \\ g'(\xi_K) \end{pmatrix} = \boldsymbol{g}_\xi = \begin{pmatrix} g_\xi(\xi_1) \\ \vdots \\ g_\xi(\xi_K) \end{pmatrix} \tag{5.8}$$

That is, the vector of solution points is $\boldsymbol{\xi}$, the vector of slope corrections at the solution points, $\boldsymbol{g}_\xi$. For the global element, set

$$\boldsymbol{x}_j = \begin{pmatrix} x_{j,1} \\ \vdots \\ x_{j,K} \end{pmatrix} \quad \text{and} \quad \boldsymbol{u}_j = \begin{pmatrix} u_{j,1} \\ \vdots \\ u_{j,K} \end{pmatrix} \tag{5.9}$$

Denote by $\boldsymbol{l}$ and $\boldsymbol{r}$ the (column) vectors of K components whose dot product with $\boldsymbol{u}_j$ yield the interface values at the left and right boundary, namely, $j-1/2$ and $j+1/2$, respectively. In the local coordinate,

$$u_j(-1) = \sum_{k=1}^{K} l_k\, u_{j,k} = \boldsymbol{l}.\boldsymbol{u}_j = \boldsymbol{l}^T\boldsymbol{u}_j \quad \text{and} \quad u_j(1) = \sum_{k=1}^{K} r_k\, u_{j,k} = \boldsymbol{r}.\boldsymbol{u}_j = \boldsymbol{r}^T\boldsymbol{u}_j \tag{5.10}$$

where the dot (.) represents dot product of two (column) vectors, the superscript T denotes the transpose, and the space (e.g., $\mathbf{l}^T\mathbf{u}_j$) represents matrix multiplication. Let $\boldsymbol{D}$ be the derivative matrix defined in (2.12) and (2.14), and let $\boldsymbol{d}_k$ be the $k-$th row of $\boldsymbol{D}$. Then

$$(u_j)_\xi(\xi_k) = \boldsymbol{d}_k\,.\boldsymbol{u}_j\,. \tag{5.11}$$

Since $f = u$ and the wind blows from the left, $f_{j-1/2,\,\mathrm{upw}} = u_{j-1}(1)$. Therefore, in the local coordinate, with F_j denoting the continuous flux function,

$$(F_j)_\xi = (u_j)_\xi + [u_{j-1}(1) - u_j(-1)]\, g_\xi = (u_j)_\xi + (\boldsymbol{r}^T\boldsymbol{u}_{j-1} - \boldsymbol{l}^T\boldsymbol{u}_j)\, g_\xi\,. \tag{5.12}$$

By evaluating the above at the solution point ξ_k and, since I has length 2,

$$\frac{du_{j,k}}{dt} = -2[\ \boldsymbol{d}_k\,.\boldsymbol{u}_j + (\boldsymbol{r}^T\boldsymbol{u}_{j-1} - \boldsymbol{l}^T\boldsymbol{u}_j)\, g_\xi(\xi_k)]\,. \tag{5.13}$$

Or, in vector form, noting that $(\boldsymbol{r}^T\boldsymbol{u}_{j-1} - \boldsymbol{l}^T\boldsymbol{u}_j)$ is a scalar quantity while $\boldsymbol{D}$ is a matrix,

$$\frac{d\boldsymbol{u}_j}{dt} = -2[\ \boldsymbol{D}\,\boldsymbol{u}_j + (\boldsymbol{r}^T\boldsymbol{u}_{j-1} - \boldsymbol{l}^T\boldsymbol{u}_j)\, \boldsymbol{g}_\xi]\,. \tag{5.14}$$

The above can be rearranged as follows. Since $\boldsymbol{l}^T\boldsymbol{u}_j$ is a scalar and $\boldsymbol{g}_\xi$ is a vector,

$$(\boldsymbol{l}^T\boldsymbol{u}_j)\boldsymbol{g}_\xi = \boldsymbol{g}_\xi(\boldsymbol{l}^T\boldsymbol{u}_j) = (\boldsymbol{g}_\xi\,\boldsymbol{l}^T)\,\boldsymbol{u}_j\,. \tag{5.15}$$

Here, $\boldsymbol{g}_\xi$ is a column vector; $\boldsymbol{l}^T$ is a row vector; as a result, $\boldsymbol{g}_\xi\,\boldsymbol{l}^T$ is a $K\times K$ matrix. Next, using (5.15), we rearrange (5.14) as

$$\frac{d\boldsymbol{u}_j}{dt} = -2\{\ (\boldsymbol{g}_\xi\,\boldsymbol{r}^T)\,\boldsymbol{u}_{j-1}\ +\ [\ \boldsymbol{D} - (\boldsymbol{g}_\xi\,\boldsymbol{l}^T)\ ]\,\boldsymbol{u}_j\ \}\,. \tag{5.16}$$

Denote ($K\times K$ matrices)

$$\boldsymbol{C}_{-1} = -2\,\boldsymbol{g}_\xi\,\boldsymbol{r}^T \qquad \text{and} \qquad \boldsymbol{C}_0 = -2[\ \boldsymbol{D} - (\boldsymbol{g}_\xi\,\boldsymbol{l}^T)\ ]\,. \tag{5.17}$$

Then, (5.16) takes the form

$$\frac{d\boldsymbol{u}_j}{dt} = \boldsymbol{C}_{-1}\,\boldsymbol{u}_{j-1}\ +\boldsymbol{C}_0\ \boldsymbol{u}_j\,. \tag{5.18}$$

Replacing $\boldsymbol{u}_{j-1}$ by $e^{-iw}\boldsymbol{u}_j$ as in (5.7), we obtain

$$\frac{d\boldsymbol{u}_j}{dt} = (e^{-iw}\boldsymbol{C}_{-1} + \boldsymbol{C}_0)\,\boldsymbol{u}_j\,.$$

Therefore, the discretization of space (semidiscretization) results in the equation (see also Chapter 10 of Hirsch 1988)

$$\frac{d\boldsymbol{u}_j}{dt} = \boldsymbol{S}\,\boldsymbol{u}_j \tag{5.19}$$

where the matrix $\boldsymbol{S}$ (for 'space' or 'semidiscrete') is defined by

$$\boldsymbol{S} = e^{-iw}\boldsymbol{C}_{-1} + \boldsymbol{C}_0 . \tag{5.20}$$

Equation (5.19) is similar to the differential equation $du/dt = \lambda u$ whose solution is $ce^{\lambda t}$ where c is an arbitrary constant. If $\mathrm{Re}(\lambda) \le 0$, the solution is stable; otherwise, it blows up as t tends to infinity. Here, the eigenvalues of $\boldsymbol{S}$ take the place of λ and, for all schemes discussed here, $\boldsymbol{S}$ has K eigenvalues. The eigenvalue which approximates $-iw$ (the exact eigenvalue of the operator $-\partial/\partial x$ given by (5.4)) is called the *principal eigenvalue* and is denoted by $S(w)$:

$$S(w) \approx -iw . \tag{5.21}$$

All other eigenvalues are spurious ones. All eigenvalues must lie in the left half of the complex plane for the semidiscretization to be stable. If any of the eigenvalues lies in the right half of the complex plane ($\mathrm{Re}(\lambda) > 0$), the scheme may eventually blow up unless the instability is mild and the time stepping provides some additional dissipation to stabilize the scheme. As will be shown below, some of the existing semi-discrete schemes are mildly unstable, but in practical calculations, the fully discrete versions are stable: the schemes convect the waves out of the domain of calculation long before any instability occurs.

To find the order of accuracy of a scheme, one can expand the principal eigenvalue $S(w)$ as a Taylor series in w. A scheme is m-th order accurate if, for small w,

$$S(w) = -iw + O(w^{m+1}) . \tag{5.22}$$

Note that S is an approximation to the scaled derivative $-u_x h$ where h is the cell width.

In practice, it is difficult if not impossible to derive an expression for $S(w)$ when the number of solution points is greater than 2. Therefore, we obtain the order of accuracy of a scheme by the following approximation. First, let w be fixed, say, $w = \pi/4$. We can estimate the error

$$E(w) = S(w) + iw . \tag{5.23}$$

Next, by halving the wave number w (which is equivalent to doubling the number of mesh points), the error corresponding to $w/2$ (say, $w/2 = \pi/8$) is

$$E(w/2) = S(w/2) + iw/2 . \tag{5.24}$$

Since

$$O\left(\left(\frac{w}{2}\right)^{m+1}\right) \approx \left(\frac{1}{2}\right)^{m+1} O(w^{m+1}) ,$$

for a scheme to be m-th order accurate,

$$E(w/2) \approx \left(\frac{1}{2}\right)^{m+1} E(w) .$$

That is,

$$\frac{E(w)}{E(w/2)} \approx 2^{m+1} .$$

Taking the logarithm of both sides of the above, the order of accuracy is given by the integer

$$m \approx \left[\mathrm{Log}\left(\frac{E(w)}{E(w/2)}\right) \Big/ \mathrm{Log}(2) \right] - 1 . \tag{5.25}$$

We now show that the eigenvalues of $\boldsymbol{S}$ defined by (5.19) and (5.20) are independent of the solution points chosen. To proceed, consider the local coordinate ξ on $I = [-1,1]$. Let ξ_k be the solution points and u_k the corresponding data. Then $\sum_{k=1}^{K} u_k \phi_k(\xi)$ is the polynomial of degree $K-1$ interpolating the data. Denote the vector $\{u_k\}_{k=1}^{K}$ by $\boldsymbol{u}$. Let $\tilde{\xi}_l$, $l = 1,...,K$ be another set of solution points, and let $\tilde{u}_l$ be the interpolated value at $\tilde{\xi}_l$,

$$\tilde{u}_l = \sum_{k=1}^{K} u_k \phi_k(\tilde{\xi}_l) \quad \text{and} \quad \tilde{\boldsymbol{u}} = \{\tilde{u}_l\}_{l=1}^{K} . \tag{5.26}$$

For $1 \le k, l \le K$, set

$$m_{lk} = \phi_k(\tilde{\xi}_l) \quad \text{and} \quad \boldsymbol{M} = \{m_{lk}\} . \tag{5.27}$$

Note that $\boldsymbol{M}$ is a $K \times K$ matrix. By (5.26) and (5.27),

$$\tilde{\boldsymbol{u}} = \boldsymbol{M}\boldsymbol{u} . \tag{5.28}$$

Since we can obtain $\boldsymbol{u}$ from $\tilde{\boldsymbol{u}}$, $\boldsymbol{M}$ is invertible. Next, equation (5.19) in the local coordinate takes the form, using the same notation as (5.19) for the sake of simplicity,

$$d\boldsymbol{u}/dt = \boldsymbol{S}\,\boldsymbol{u} . \tag{5.29}$$

For the solution points $\tilde{\xi}_l$, the corresponding equation is

$$d\tilde{\boldsymbol{u}}/dt = \tilde{\boldsymbol{S}}\,\tilde{\boldsymbol{u}} . \tag{5.30}$$

We wish to show that $\tilde{\boldsymbol{S}}$ and $\boldsymbol{S}$ are similar, and thus, share the same eigenvalues. To this end, multiplying (5.29) on the left by $\boldsymbol{M}$, we have

$$\boldsymbol{M}\,d\boldsymbol{u}/dt = \boldsymbol{M}\boldsymbol{S}\,\boldsymbol{u} . \tag{5.31}$$

Therefore, since $\boldsymbol{M}$ is invertible and independent of t,

$$d(\boldsymbol{M}\,\boldsymbol{u})/dt = \boldsymbol{M}\boldsymbol{S}\boldsymbol{M}^{-1}\boldsymbol{M}\,\boldsymbol{u} = (\boldsymbol{M}\boldsymbol{S}\boldsymbol{M}^{-1})(\boldsymbol{M}\,\boldsymbol{u}) . \tag{5.32}$$

That is, by (5.28),

$$d\tilde{\boldsymbol{u}}/dt = (\boldsymbol{M}\boldsymbol{S}\boldsymbol{M}^{-1})\,\tilde{\boldsymbol{u}} . \tag{5.33}$$

Comparing (5.30) and the above, we obtain

$$\tilde{\boldsymbol{S}} = \boldsymbol{M}\boldsymbol{S}\boldsymbol{M}^{-1} . \tag{5.34}$$

That is, $\tilde{\boldsymbol{S}}$ is similar to $\boldsymbol{S}$, and the proof is completed.

Note that the derivative matrix follows the same transformation rule, namely, $\tilde{\boldsymbol{D}} = \boldsymbol{M}\boldsymbol{D}\boldsymbol{M}^{-1}$. As for $\boldsymbol{l}^T$ defined by (5.10), the transformation rule is $\tilde{\boldsymbol{l}}^T = \boldsymbol{l}^T\boldsymbol{M}^{-1}$, whereas for $\boldsymbol{g}_\xi$ in (5.8), $\tilde{\boldsymbol{g}}_\xi = \boldsymbol{M}\,\boldsymbol{g}_\xi$. We omit the details.

We conclude this section with a discussion on eigenvectors and superaccuracy. Assume that the initial condition is $u_{\text{init}}(x) = e^{iwx}$, or if we wish to have a real function, $u_{\text{init}}(x) = \cos(wx) = \text{Re}(e^{iwx})$. To start the calculation, we define $\boldsymbol{u}_j = \{u_{j,k},\, k = 1,...,K\}$ by evaluating at the solution points,

$$\boldsymbol{u}_j = \{\exp[i\,w(j+\xi_k/2)]\}_{k=1}^K . \tag{5.35}$$

For each cell E_j, the mapping from $u_{\text{init}}(x)$ to $u_j(x) = \sum u_{j,k}\,\phi_{j,k}(x)$ can be considered as a 'projection' of the initial condition onto $\boldsymbol{P}_{K-1}$. The error $u_{\text{init}}(x) - u_j(x)$ is of order $O(w^K)$ (i.e., $O(h^K)$). Thus, if we calculate the error by comparing the solution $u_j(x)$ at the final time with u_{exact} defined in (5.3), the scheme is accurate to order no higher than K.

One way to observe the property of super-convergence or super-accuracy numerically is as follows. Let $\boldsymbol{S}$ be the matrix representing the spatial discretization defined by (5.20). For the schemes discussed here, the eigenvectors of $\boldsymbol{S}$ denoted by $\boldsymbol{e}_k$ are independent: $\{\boldsymbol{e}_k\}_{k=1}^K$ form a basis for $\boldsymbol{R}^K$. Let $\boldsymbol{e}_1$ be the principal eigenvector corresponding to the principal eigenvalue $S(w)$ (a more precise notation is $\boldsymbol{e}_1(w)$). Since $\boldsymbol{u}_j = e^{iwj}\boldsymbol{u}_0$, we only need to focus on $j = 0$. The vector $\boldsymbol{u}_0$ can be decomposed using the basis $\{\boldsymbol{e}_k\}_{k=1}^K$ by

$$\boldsymbol{u}_0 = \sum\nolimits_{k=1}^K c_k\boldsymbol{e}_k . \tag{5.36}$$

The principal component is $c_1\boldsymbol{e}_1$, which corresponds to the principal eigenvalue $S(w)$. This component is propagated with the accuracy of the scheme, i.e., of order m as discussed above if the scheme is (super) accurate to order m. For this component, the error accumulates with the number of time steps. Thus, one way to observe the super-convergence or super-accuracy property is to use the principal eigenvalue $e^{ijw}c_1\boldsymbol{e}_1$ as start-up solution and calculate the error by comparing the final solution with this start-up solution.

VI. Stability and Accuracy of Schemes via Flux Reconstruction

As discussed in the previous section, Fourier stability and accuracy analyses of schemes are independent of the solution points chosen. In other words, whether the Gauss or Radau or Lobatto or equidistant points are selected for the solution points, the results from Fourier analysis are the same. These results vary only with the choice of

correction function. For brevity, we often identify the scheme by its correction function, e.g., scheme g_{Ga} refers to the semidiscrete flux reconstruction scheme corresponding to the correction function g_{Ga}.

The case $K=1$. The corresponding correction function is $g=(1-\xi)/2$. Equation (5.18) then takes the form of the first order upwind scheme:

$$\frac{du_j}{dt}=u_{j-1}-u_j.$$

The eigenvalue is

$$S(w)=e^{-iw}-1=-iw-w^2/2+O(w^3).$$

As w varies on $[0,2\pi]$, $S(w)$ varies on a circle of radius 1 centered at $-1+0i$ on the complex plane, which is called the spectrum of the semidiscrete scheme.

Note that for this correction function, $\int_{-1}^{1} g(\xi)d\xi=1$; thus, $\int_{E_j} f_j(x)dx \neq \int_{E_j} F_j(x)dx$.

The case $K=2$. The correction functions are shown in Fig. 6.1(a). They are

$$g_{\mathrm{DG}}=\frac{3\xi^2}{4}-\frac{\xi}{2}-\frac{1}{4},\quad g_{\mathrm{SG}}=g_{\mathrm{Lo}}=g_{\mathrm{Ga}}=\frac{\xi^2}{2}-\frac{\xi}{2},\quad \text{and}\quad g_2=g_{\mathrm{Lump,\,Lo}}=\frac{\xi^2}{4}-\frac{\xi}{2}+\frac{1}{4}.$$

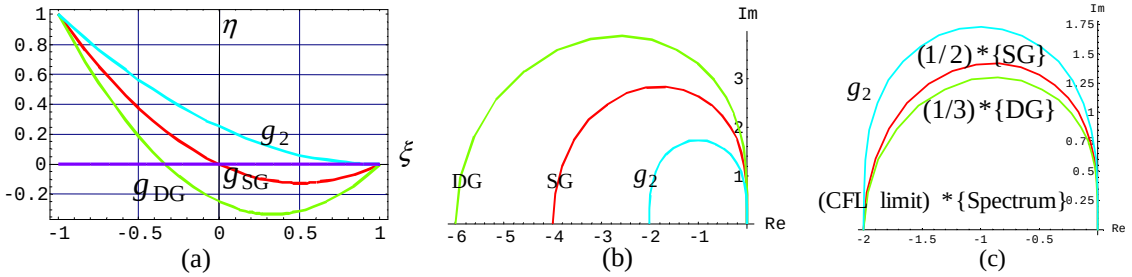


Figure 6.1. The case $K=2$. **(a) Correction functions**. **(b) Spectra of semi-discrete schemes** *(collection of all eigenvalues). Due to symmetry, only the top half of the complex plane is shown. The spectrum of scheme* g_2 *turns out to be identical to the boundary of the stability region of the explicit 2-stage Runge-Kutta time stepping method. The maximum (artificial) dissipation occurs at* $x_2=-2$ *for scheme* g_2 *(lumping for Lobatto points),* $x_{\mathrm{SG}}=-4$ *for staggered-grid, and* $x_{\mathrm{DG}}=-6$ *for DG.* **(c) Plots of (CFL limit)*spectrum**. *If we use explicit 2-stage Runge-Kutta time stepping, the CFL limit for DG scheme is* $x_2/x_{\mathrm{DG}}=1/3$*; that for SG,* $x_2/x_{\mathrm{SG}}=1/2$*; and for scheme* g_2, 1. *Note that the curves (spectrum* CFL limit) lie inside the stability region of the 2-stage R-K scheme, which is the region bounded by curve* g_2.

($K=2$) Scheme	Order of accuracy	Coarse mesh error, $w=\pi/8$	Fine mesh error, $w=\pi/16$
DG	3	$-3.2\times10^{-4}-3.3\times10^{-5}i$	$-2.1\times10^{-5}-1.1\times10^{-6}i$
SG	2	$-7.1\times10^{-4}+2.4\times10^{-3}i$	$-4.6\times10^{-5}+3.1\times10^{-4}i$
$g_2=g_{\mathrm{Lump,Lo}}$	2	$-2.5\times10^{-3}+9.\times10^{-3}i$	$-7.1\times10^{-4}+2.4\times10^{-3}i$

Table 6.1. Orders of accuracy and errors of schemes for $K=2$. *All errors have negative real parts. A closer inspection shows that all eigenvalues lie on the left half of the complex plane; as a result, these schemes are stable.*

The spectra (collection of all eigenvalues) of the corresponding semidiscrete schemes are shown in Fig. 6.1(b). Due to symmetry, only the top half of the complex plane is shown. Each spectrum intersect the real axis at a location corresponding to the maximum numerical dissipation of the scheme: at $x_2=-2$ for g_2, $x_{\mathrm{SG}}=-4$ for the staggered-grid, and $x_{\mathrm{DG}}=-6$ for DG scheme. These locations determine the CFL limits (the maximum $|a|\Delta t/\Delta x$ before the scheme blows up, where a is the wave speed, Δt time step size, and Δx mesh width).

Note the following surprising coincidence. For $K=2$, the spectrum of the scheme g_2 (spatial discretization) turns out to be identical to the boundary of the stability region of the explicit 2-stage Runge-Kutta time stepping method. This coincidence holds true for all K tested by this author: the spectrum of the scheme g_K turns out to be identical to the boundary of the stability region of the explicit time stepping method with amplification factor

$1+z+\ldots+z^K$ where z is a complex number. (The K-stage Runge-Kutta method has such an amplification factor for $K \le 4$.)

If the 2-stage Runge-Kutta time stepping is employed, then the CFL limit equals 1 for scheme g_2; 1/2 for SG; and 1/3 for DG. In addition, the curves (CFL limit) *spectrum shown in Fig 6.1(c) lie inside the region bounded by curve g_2.

The orders of accuracy and errors of the semi-discrete schemes for $K = 2$ are tabulated in Table 6.1. Observe that the errors for all schemes have negative real parts, i.e., the eigenvalues lie on the left half of the complex plane. A closer inspection shows that all eigenvalues share this property; therefore, these schemes are stable.

The case $K = 3$. The correction functions are shown in Fig. 6.2 where $g_1 = g_{DG}$, $g_2 = g_{Lump,\,Lo}$, and g_3 are shown in (a) and g_{SG}, g_{Lo}, and g_{Ga}, in (b). Note that for each of the latter three, the interior zeros are symmetric about 0. Also note that for this K, $g_{Lump,\,Ch\text{-}Lo} = g_{Lump,\,Lo} = g_2$.

For $K = 3$, the spectra (collection of all eigenvalues) of the corresponding semidiscrete schemes are shown in Fig. 6.3(a). Similar to the case $K = 2$, note that the spectrum of scheme g_3 turns out to be identical to the boundary of the stability region of the explicit 3-stage Runge-Kutta method.

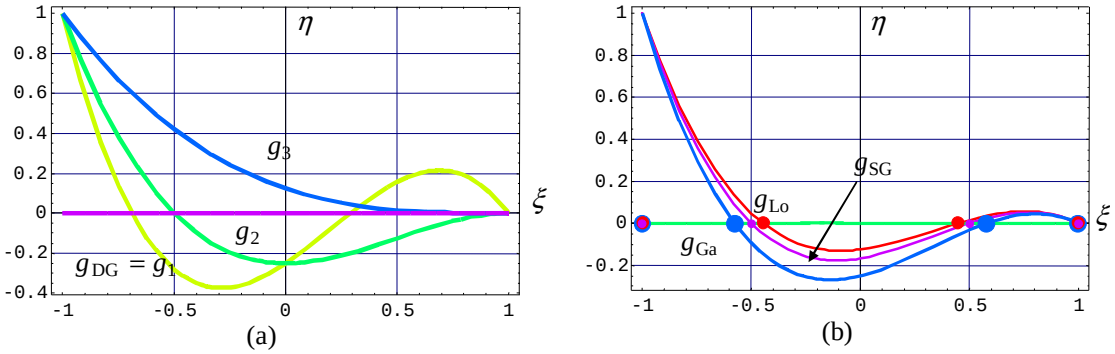


Figure 6.2. Correction functions for the case $K = 3$; **(a)** $g_1 = g_{DG}$, $g_2 = g_{Lump,\,Lo}$ *(lumping for Lobatto points)*, and g_3; **(b)** g_{SG}, g_{Lo}, *and* g_{Ga}; *the interior zeros for each of these three correction functions (dots) are symmetric about* 0.

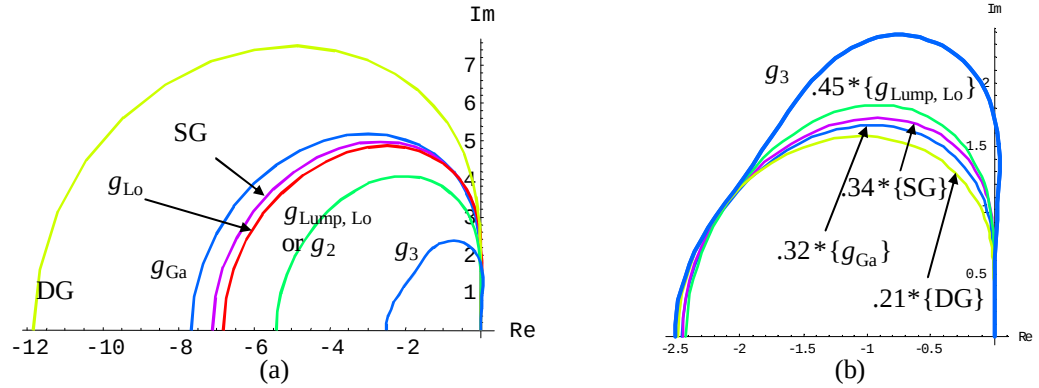


Figure 6.3. (a) Spectra of semidiscrete schemes for the case $K = 3$. **(b) Plots of (CFL limit)*spectrum**. *If we use the explicit 3-stage Runga-Kutta time stepping for these schemes, the CFL limit for the DG scheme is about .21 (slightly smaller than* x_3 / x_{DG}*); for SG, .34; and for* g_2, .45*. Note that the curves (CFL limit) *spectrum lie inside the region bounded by curve* g_3*, which is identical to the stability region of the 3-stage Runge-Kutta scheme.*

Next, we discuss CFL limits. First, as shown in Fig. 6.3, the spectra intersect the real axis at: $x_{DG} \approx -11.8$ for DG, $x_{SG} \approx -7.1$ for staggered-grid, $x_2 \approx -5.4$ for scheme g_2 (lumping for Lobatto points), and $x_3 \approx -2.5$ for scheme g_3. If the 3-stage Runge-Kutta time stepping is employed, then for DG, the CFL limit is (slightly smaller

than x_3 / x_{DG}) about .21, SG .34, g_{Ga} .32, and $g_{Lump,Lo}$ .45. Fig. 6.3(b) shows the plots of (CFL limit)*spectrum for the schemes DG, SG, $g_{Lump,Lo}$, and g_{Ga} . Note that these curves lie inside the region bounded by the curve g_3 , which is identical to the stability region of the 3-stage Runge-Kutta time stepping scheme.

($K=3$) Scheme	Order of accuracy	Coarse mesh error, $w=\pi/8$	Fine mesh error, $w=\pi/16$
DG	5	$-5.\times10^{-7}-3.4\times10^{-8}i$	$-7.9\times10^{-9}-2.7\times10^{-10}i$
g_{Ga}	4	$-1.4\times10^{-6}+8.5\times10^{-6}i$	$-2.2\times10^{-8}+2.7\times10^{-7}i$
$g_2 = g_{Lump,Lo}$	4	$-3.2\times10^{-6}+1.9\times10^{-5}i$	$-5.\times10^{-8}+6.\times10^{-7}i$
SG	3	$3.9\times10^{-5}+1.4\times10^{-5}i$	$2.5\times10^{-6}+4.4\times10^{-7}i$
g_{Lo}	3	$6.4\times10^{-5}+1.7\times10^{-5}i$	$4.1\times10^{-6}+5.4\times10^{-7}i$
g_3	3	$9.3\times10^{-4}+3.1\times10^{-4}i$	$6.1\times10^{-5}+9.8\times10^{-6}i$

Table 6.2. Orders of accuracy and errors of schemes with 3 solution points ($K=3$). *The errors for schemes DG,* g_{Ga} *, and* $g_{Lu,Lo}$ *have negative real parts. A closer inspection of all eigenvalues shows that they lie on the left half of the complex plane, and these three schemes are stable. On the other hand, errors for the three schemes* g_{SG} *,* g_{Lo} *, and* g_3 *have positive real parts; thus, these schemes are unstable; for* g_{SG} *and* g_{Lo} *, however, the instability is mild.*

The orders of accuracy and errors of the semi-discrete schemes for $K=3$ are tabulated in Table 6.2. The errors for schemes DG, g_{Ga} , and $g_{Lu,Lo}$ have negative real parts; a closer check of the eigenvalues reveals that these three schemes are stable. On the other hand, errors for the three schemes g_{SG} , g_{Lo} , and g_3 have positive real parts; thus, these schemes are unstable; for the schemes g_{SG} and g_{Lo} , however, the instability is mild.

The next remark holds for all schemes and arbitrary K . Between the real and imaginary parts of the errors, if the order of accuracy is odd, the leading error is real; otherwise, it is imaginary.

Finally, for the staggered-grid scheme, the maximum value of the real parts of the eigenvalues is about 0.0028, which shows that the instability is mild. For this scheme, the unstable component is usually convected out of the domain of calculation before it has a chance to contaminate the solution.

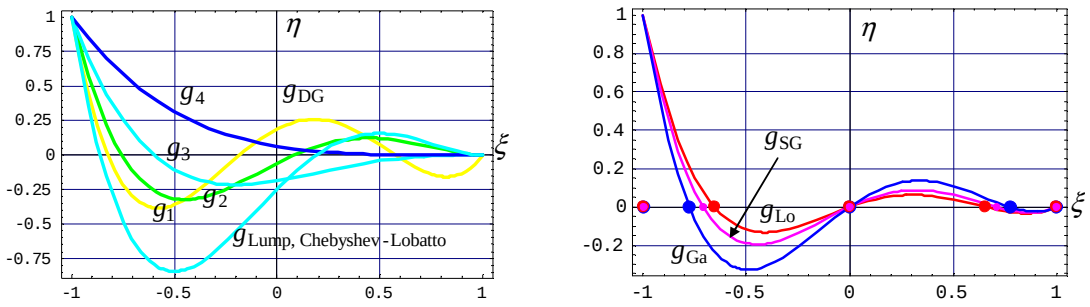


Figure 6.4. Correction functions for the case $K=4$; **(a)** $g_1 = g_{DG}$, $g_2 = g_{Lump,Lo}$, g_3 , g_4 , *and* $g_{Lump,Ch-Lo}$ *(lumping for Chebyshev-Lobatto points);* **(b)** g_{SG} , g_{Lo} , *and* g_{Ga} ; *the interior zeros for each of these three correction functions (dots) are symmetric about* 0.

The case $K=4$. The correction functions are shown in Fig. 6.4 where $g_1 = g_{DG}$, $g_2 = g_{Lump,Lo}$, g_3 , g_4 and $g_{Lump,Ch-Lo} = g_{Lump,Chebyshev-Lobatto}$ are shown in (a) and g_{SG} , g_{Lo} , and g_{Ga} , in (b). For each of the last three, the interior zeros are symmetric about 0.

For $K=4$, the spectra (collection of all eigenvalues) of the corresponding semidiscrete schemes are shown in Fig. 6.5(a). Similar to the cases of $K=2$ and $K=3$, the spectrum of scheme g_4 is identical to the boundary of the stability region of the explicit 4-stage Runge-Kutta method.

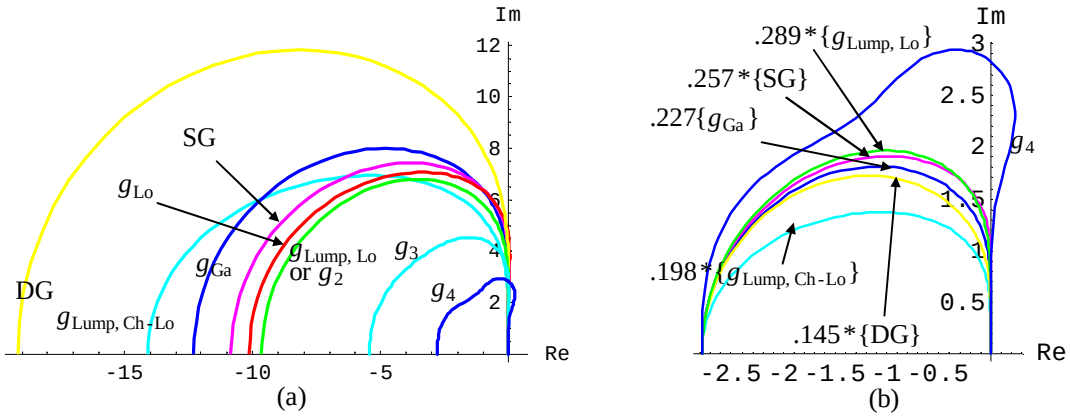


Figure 6.5. (a) Spectra of semidiscrete schemes with $K = 4$. (b) Plots of (CFL limit)*spectrum. *If we use the explicit 4-stage Runge-Kutta time stepping for these schemes, the CFL limit for the scheme DG is about .21; for SG, .34; and for g_2, .45. Note that the curves (CFL limit)*spectrum lie inside the stability region of the 4-stage R-K scheme, which is the region bounded by curve g_4.*

Next, we discuss CFL limits. First, the spectra intersect the real axis at: $x_{DG} \approx -19.2$ for DG, $x_{Lump, Ch-Lo} \approx -14.1$ for scheme $g_{Lump, Ch-Lo}$ (lumping for Chebyshev-Lobatto points), $x_{Ga} \approx -12.3$ for g_{Ga}, $x_{SG} \approx -10.8$ for staggered-grid, $x_2 \approx -9.6$ for g_2 (lumping for Lobatto points), and $x_4 \approx -2.8$ for g_4. If the 4-stage Runge-Kutta time stepping is employed, then in increasing order, the CFL limits are, for DG, $x_4 / x_{DG} \approx .145$; for $g_{Lump, Ch-Lo}$, .198; for g_{Ga}, .227; for SG, $x_4 / x_{SG} \approx .257$; and for $g_{Lump, Lo}$, .289. Thus, the CFL limit for $g_{Lump, Lo}$ scheme is about twice that for DG for the case $K = 4$.

Fig. 6.5(b) shows the plots of (CFL limit)*spectrum for the following schemes: DG, SG, $g_{Lump, Lo}$, $g_{Lump, Ch-Lo}$, and g_{Ga}. Note that these curves lie inside the stability region of the 4-stage Runge-Kutta time stepping scheme.

($K = 4$) Scheme	Order of accuracy	Coarse mesh error, $w = \pi/4$	Fine mesh error, $w = \pi/8$
DG	7	$-1.\times10^{-7} - 1.\times10^{-8} i$	$-4.\times10^{-10} - 2.\times10^{-11} i$
g_{Ga}	6	$-3.1\times10^{-7} + 1.3\times10^{-6} i$	$-1.2\times10^{-9} + 1.1\times10^{-8} i$
$g_2 = g_{Lump, Lo}$	6	$-5.4\times10^{-7} + 2.3\times10^{-6} i$	$-2.2\times10^{-9} + 1.9\times10^{-8} i$
$g_{Lump, Ch-Lo}$	4	$-3.3\times10^{-5} + 9.8\times10^{-5} i$	$-5.3\times10^{-7} + 3.2\times10^{-6} i$
SG	4	$1.2\times10^{-5} - 2.2\times10^{-5} i$	$2.\times10^{-7} - 7.8\times10^{-7} i$
g_3	5	$5.9\times10^{-5} + 2.5\times10^{-5} i$	$9.9\times10^{-7} + 2.\times10^{-7} i$
g_{Lo}	4	$2.1\times10^{-5} - 3.9\times10^{-5} i$	$3.4\times10^{-7} - 1.3\times10^{-6} i$
g_4	4	$1.5\times10^{-3} - 2.\times10^{-3} i$	$2.5\times10^{-5} - 7.4\times10^{-5} i$

Table 6.3. Orders of accuracy and errors of schemes for $K = 4$. *The errors for schemes DG, g_{Ga}, g_2 (lumping for Lobatto points), and $g_{Lump, Ch-Lo}$ (lumping for Chebyshev-Lobatto points) have negative real parts; these four schemes are stable. On the other hand, errors for the four schemes SG, g_3 g_{Lo}, and g_4 have positive real parts; these schemes are unstable, but the instability of schemes SG and g_3 are mild.*

The orders of accuracy and errors of the semi-discrete schemes for $K = 4$ are tabulated in Table 6.3. The errors for schemes DG, g_{Ga}, g_2 (lumping for Lobatto points), and $g_{Lump, Ch-Lo}$ (lumping for Chebyshev-Lobatto points), have negative real parts. A closer check of all eigenvalues reveals that these four schemes are stable. The errors for the last four schemes, namely, SG, g_3, g_{Lo}, and g_4 have positive real parts; as a result, they are unstable. Among

these four, the staggered-grid scheme has the mildest instability and the smallest error, whereas the scheme g_4 has the most severe instability and the largest error. For this reason, if g_4 were to be employed, it must be blended with other stable schemes, possibly in a manner similar to the use of unstable stencils in the ENO and WENO methods.

Finally, for each scheme, the real parts of the eigenvalues reach a maximum: for the staggered-grid scheme, this maximum is about 0.015; for g_3, 0.011; for g_{Lo}, 0.050, and for g_4, 0.237.

The case $K=8$. The correction functions are shown in Fig. 6.6; $g_1 = g_{\text{DG}}$, $g_2 = g_{\text{Lump, Lo}}$, g_3, and $g_{\text{Lump, Ch-Lo}} = g_{\text{Lump, Chebyshev-Lobatto}}$ in (a); g_{SG} and g_{Ga} in (b). For the last two correction functions, the interior zeros are symmetric about 0.

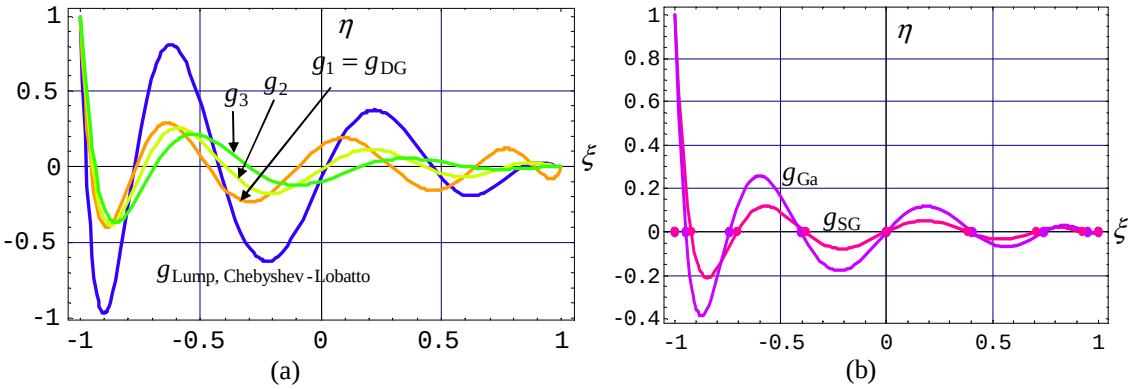


Figure 6.6. Correction functions for $K=8$; **(a)** $g_1 = g_{\text{DG}}$, $g_2 = g_{\text{Lump, Lo}}$, g_3, and $g_{\text{Lump, Ch-Lo}}$; **(b)** g_{SG} *and* g_{Ga}.

The spectra of the semidiscrete schemes for $K=8$ are shown in Fig. 6.7.

The orders of accuracy and errors of the semi-discrete schemes for $K=8$ are tabulated in Table 6.4. Due to round-off errors, numbers less than 10^{-12} are meaningless. The errors for schemes DG, g_{Ga}, g_2 (lumping for Lobatto points), and $g_{\text{Lump, Ch-Lo}}$ (lumping for Chebyshev-Lobatto points) have negative real parts; these four schemes are stable. The errors for schemes SG and g_3 have positive real parts; these schemes are mildly unstable. The maximum value of the real parts of the eigenvalues is 0.000040 for SG and 0.00011 for g_3.

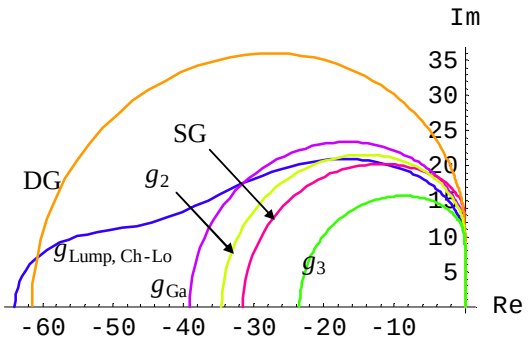


Figure 6.7. Spectra of schemes with $K=8$.

Remaks for arbitrary K.

Concerning the instability of the staggered grid schemes, the maximum value of the real parts of the eigenvalues are listed for $K \le 12$ in Table 6.5. Note that the larger the value, the more unstable the scheme. The scheme corresponding to $K=8$ is least unstable: the maximum value is only 0.000041.

The schemes DG, g_{Ga}, and $g_{\text{Lump, Lo}}$ are stable for all values K tested by this author. Thus, it appears that if g is perpendicular to $\boldsymbol{P}_{K-3}$, then the resulting scheme is stable. The inverse, however, does not hold: the scheme $g_{\text{Lump, Ch-Lo}}$ is not perpendicular to any $\boldsymbol{P}_m$, and it is stable. The following problems remain open: (1) What are the necessary and sufficient conditions on the correction function for the resulting scheme to be stable? (2) Can a scheme be optimized in a manner that it is mildly unstable (say the maximum real parts of the eigenvalues are less than 0.001) but the CFL limit is largest possible?

Concerning the relation between the correction function and super-accuracy, from the above Fourier analysis, if g is perpendicular to $\boldsymbol{P}_m$, then the resulting scheme is super-accurate to order $K+m+1$.

($K=8$) Scheme	Order of accuracy	Coarse mesh error, $w=.9\pi$	Fine mesh error, $w=.45\pi$
DG	15?	$-1.1\times10^{-10}-1.9\times10^{-11}i$	$\lvert\text{error}\rvert<10^{-12}$
g_{Ga}	14?	$-3.8\times10^{-10}+9.4\times10^{-10}i$	$\lvert\text{error}\rvert<10^{-12}$
$g_2=g_{\text{Lump, Lo}}$	14?	$-4.9\times10^{-10}+1.2\times10^{-9}i$	$\lvert\text{error}\rvert<10^{-12}$
$g_{\text{Lump, Ch-Lo}}$	8	$-4.3\times10^{-7}+2.2\times10^{-7}i$	$-6.8\times10^{-10}+1.3\times10^{-9}i$
SG	8	$1.\times10^{-6}+4.4\times10^{-8}i$	$1.5\times10^{-9}-1.5\times10^{-9}i$
g_3	13?	$1.3\times10^{-8}+9.3\times10^{-9}i$	$\lvert\text{error}\rvert<10^{-12}$

Table 6.4. Orders of accuracy and errors of schemes for $K=8$. *Note that due to round-off errors, numbers less than* 10^{-12} *are meaningless. The errors for schemes DG,* g_{Ga}, g_2 *(lumping for Lobatto points), and* $g_{\text{Lump, Ch-Lo}}$ *(lumping for Chebyshev-Lobatto points) have negative real parts; these four schemes are stable. The errors for schemes SG and* g_3 *have positive real parts; these schemes are mildly unstable. The maximum positive real parts are: 0.000040 for SG and 0.00011 for* g_3.

K	3	4	5	6	7	8	9	10	11	12
Max. Value	0.028	0.015	0.028	0.031	0.011	0.000041	0.00037	0.0016	0.0041	0.0082

Table 6.5. The maximum value of the real parts of the eigenvalues for the staggered-grid schemes with various K**.**

VII. Flux Reconstruction and DG Approaches

DG scheme via Flux Reconstruction Approach.

Next, we prove that the DG method can be cast as a flux reconstruction scheme: it is identical to the scheme using the Radau polynomials as correction functions. To this end, we first rederive the DG scheme for the conservation law $u_t+f_x=0$.

On the cell (or element) $E=E_j$, let the inner product of any two functions v and w be defined by

$$(v,w)=\int_E v(x)w(x)\,dx. \tag{7.1}$$

Formally, we require the solution u to satisfy, for any test function (polynomial of degree $K-1$) ϕ,

$$\frac{\partial}{\partial t}\int_E u\,\phi\,dx+\int_E f_x\,\phi\,dx=0. \tag{7.2}$$

Applying integration by parts to the second term above, we obtain

$$\frac{\partial}{\partial t}\int_E u\,\phi\,dx+\left[f\,\phi\right]_{\partial E}-\int_E f\,\phi_x\,dx=0. \tag{7.3}$$

For the DG method, the function u is replaced by the polynomial $u_j(x)$ of degree $K-1$, ϕ by one of the basis functions $\phi_{j,k}$, $k=1,...,K$, and f by the piecewise polynomial $f_j(x)$ also of degree $K-1$. As for the boundary term, if $E=E_j=[x_{j-1/2},x_{j+1/2}]$, then $[f\,\phi]_{\partial E}=(f\,\phi)(x_{j+1/2})-(f\,\phi)(x_{j-1/2})$. Since the piecewise polynomial function $\{u_j\}$ is discontinuous across cell interfaces, so is the piecewise polynomial $\{f_j\}$. Which value of f should we employ to evaluate the boundary term? It is here where the upwind fluxes $f_{j-1/2,\text{upw}}$ and $f_{j+1/2,\text{upw}}$ play their role. The value of ϕ at the boundary, however, is that from the cell E (i.e., no upwinding). Again, the upwind flux provides coupling (interaction) between the unknowns in adjacent cells and results in a conservative scheme. Denoting the upwind flux by f_{upw}, the solution u is required to satisfy

$$\frac{\partial}{\partial t}\int_E u\,\phi\,dx+\left[f_{\text{upw}}\,\phi\right]_{\partial E}-\int_E f\,\phi_x\,dx=0. \tag{7.4}$$

For the DG scheme, the solution is obtained by evaluating the above numerically. Here, however, we wish to eliminate the test function ϕ. Applying integration by parts once more to $\int_E f\,\phi_x\,dx$, since (the polynomials) f and ϕ are smooth on E,

$$\int_E f\,\phi_x\,dx = [f\,\phi]_{\partial E} - \int_E f_x\,\phi\,dx. \tag{7.5}$$

For the boundary term, in this case, both ϕ and f are calculated from the data inside the cell E (i.e., no upwinding). Substitute this expression into (7.4), we obtain,

$$\frac{\partial}{\partial t}\int_E u\phi\,dx + [(f_{\text{upw}} - f)\phi]_{\partial E} + \int_E f_x\,\phi\,dx = 0. \tag{7.6}$$

Note that $(f_{\text{upw}} - f)$ in the boundary term is simply the jump at the interface in (3.4). The above, loosely put, is the result of integrating by parts twice. It is simply the inner product of the conservation law $u_t + f_x = 0$ with the test function ϕ on E, except for the correction term $[(f_{\text{upw}} - f)\phi]_{\partial E}$. This term takes into account the upwind effect (the interaction of the data between adjacent cells).

Next, we switch to the local coordinate ξ on $I = [-1,1]$. Denote the length of E by h. Noting that $dx = (h/2)d\xi$ and $f_x\,dx = f_\xi\,d\xi$, in the local coordinate, Eq. (7.6) takes the form

$$\frac{h}{2}\frac{\partial}{\partial t}\int_{-1}^{1} u\phi\,d\xi + [\,f_{\text{upw}}(1) - f(1)\,]\phi(1) - [\,f_{\text{upw}}(-1) - f(-1)\,]\,\phi(-1) + \int_{-1}^{1} f_\xi\,\phi\,d\xi = 0. \tag{7.7}$$

The next technique is crucial since it helps eliminate the test function ϕ. Focusing first on the left boundary $\xi = -1$, in view of the expression $\int_{-1}^{1} f_\xi\,\phi\,d\xi$ above, the following question can be raised: can we find a polynomial g_{LB} on $I = [-1,1]$ which possesses the property that for any polynomial ϕ of degree $K-1$ or less,

$$-\phi(-1) = \int_{-1}^{1} g_{\text{LB}}'\,\phi\,d\xi. \tag{7.8}$$

Here, $g_{\text{LB}}'(\xi) = (g_{\text{LB}})_\xi(\xi)$. From (7.7), we require $g_{\text{LB}}'(\xi)$ to have the same degree as $u_j(\xi)$, i.e., of degree $K-1$; as a result, g_{LB} is required to be of degree K. Applying integration by parts to the above right hand side, we have

$$-\phi(-1) = g_{\text{LB}}(1)\,\phi(1) - g_{\text{LB}}(-1)\,\phi(-1) - \int_{-1}^{1} g_{\text{LB}}\,\phi'\,d\xi. \tag{7.9}$$

Thus, Eq. (7.8) holds if g_{LB} satisfies,

$$g_{\text{LB}}(-1) = 1, \qquad g_{\text{LB}}(1) = 0, \tag{7.10}$$

and, for all ϕ in $\boldsymbol{P}_{K-1}$,

$$\int_{-1}^{1} g_{\text{LB}}\,\phi'\,d\xi = 0. \tag{7.11}$$

Since ϕ is of degree $K-1$, ϕ' is of degree $K-2$; moreover, ϕ' spans $\boldsymbol{P}_{K-2}$ as ϕ spans $\boldsymbol{P}_{K-1}$. The above then implies that g_{LB} is orthogonal to $\boldsymbol{P}_{K-2}$, i.e., for any polynomial φ of degree $K-2$,

$$\int_{-1}^{1} g_{\text{LB}}\,\varphi\,d\xi = 0. \tag{7.12}$$

The criterion that g_{LB} is orthogonal to $\boldsymbol{P}_{K-2}$ provides $K-1$ conditions in determining g_{LB}. The other two are provided by (7.10). These $K+1$ conditions imply that g_{LB} is the right Radau polynomial on I (see the Appendix):

$$g_{\text{LB}}(\xi) = R_R(K,\xi). \tag{7.13}$$

Again, note that the correction function for the left boundary is the right Radau polynomial, which vanishes at the right boundary $\xi = 1$.

Another way to look at g_{LB} defined by (7.8) is by using distribution theory. Equation (7.8) implies that $-g_{\text{LB}}'$ is an approximation to the Dirac delta function at $\xi = -1$:

$$-g_{\mathrm{LB}}{}' \approx \delta(-1)\,.$$

Since the integral of the Dirac delta function is the unitstep (or Heaviside) function,

$$-g_{\mathrm{LB}} + \text{constant} \approx H(\xi+1) = \begin{cases} 0 & \text{if} \quad \xi \le -1, \\ 1 & \text{otherwise.} \end{cases}$$

By choosing the constant to be 1, after a little algebra, the above implies that g_{LB} approximates the 'step-down function' h_{-1} defined on I by

$$h_{-1}(\xi) = \begin{cases} 1 & \text{if} \quad \xi = -1, \\ 0 & \text{if} \quad \xi > -1. \end{cases} \tag{7.14}$$

Now, we switch to the right boundary. In a manner similar to (7.10) and (7.12), let g_{RB} be a polynomial of degree K defined by

$$g_{\mathrm{RB}}(-1) = 0, \qquad g_{\mathrm{RB}}(1) = 1\,, \tag{7.15}$$

and, for all φ in $\boldsymbol{P}_{K-2}$,

$$\int_{-1}^{1} g_{\mathrm{RB}}\, \varphi\, d\xi = 0\,. \tag{7.16}$$

Then g_{RB} is left Radau polynomial, which vanishes at the left boundary $\xi = -1$,

$$g_{\mathrm{RB}}(\xi) = R_L(K, \xi)\,. \tag{7.17}$$

Let the 'step-up' function h_1 be defined on I by

$$h_1(\xi) = \begin{cases} 0 & \text{if} \quad \xi < 1, \\ 1 & \text{if} \quad \xi = 1. \end{cases} \tag{7.18}$$

Then g_{RB} approximates h_1. For any ϕ in $\boldsymbol{P}_{K-1}$, ϕ' is in $\boldsymbol{P}_{K-2}$. Replacing φ by ϕ' for the left hand side of (7.16) and applying integration by parts, we obtain

$$\int_{-1}^{1} g_{\mathrm{RB}}\, \phi'\, d\xi = g_{\mathrm{RB}}(1)\phi(1) - g_{\mathrm{RB}}(-1)\phi(-1) - \int_{-1}^{1} g_{\mathrm{RB}}{}'\, \phi\, d\xi\,. \tag{7.19}$$

Using (7.15) and (7.16), the above implies

$$\phi(1) = \int_{-1}^{1} g_{\mathrm{RB}}{}'\, \phi\, d\xi\,. \tag{7.20}$$

We now return to our main task of eliminating ϕ. By (7.20) and (7.8), Eq. (7.7) can be written as

$$\frac{h}{2}\frac{\partial}{\partial t}\int_I u\phi\, d\xi + \left[f_{\mathrm{upw}}(1) - f(1)\right]\int_I g_{\mathrm{RB}}{}'\phi\, d\xi + \left[f_{\mathrm{upw}}(-1) - f(-1)\right]\int_I g_{\mathrm{LB}}{}'\phi\, d\xi + \int_I f_\xi \phi\, d\xi = 0\,. \tag{7.21}$$

What is crucial here is that we can factor ϕ out:

$$\frac{h}{2}\frac{\partial}{\partial t}\int_I u\phi\, d\xi + \int_I \left\{ f + [\,f_{\mathrm{upw}}(1) - f(1)]\, g_{\mathrm{RB}} + [\,f_{\mathrm{upw}}(-1) - f(-1)]\, g_{\mathrm{LB}} \right\}_\xi \phi\, d\xi = 0\,. \tag{7.22}$$

Thus, if we approximate u by u_j and f by f_j which are polynomials of degree $K-1$, and if we set

$$F_j(\xi) = f_j(\xi) + [\,f_{j+1/2,\,\mathrm{upw}}(1) - f_j(1)]\, g_{\mathrm{RB}}(\xi) + [\,f_{j-1/2,\,\mathrm{upw}}(-1) - f_j(-1)]\, g_{\mathrm{LB}}(\xi) \tag{7.23}$$

then, Eq. (7.22) can be written as

$$\int_I [\,(u_j)_t + \frac{2}{h_j}(F_j)_\xi\,]\ \phi\, d\xi = 0\,. \tag{7.24}$$

Switching to the global coordinate,

$$\int_{E_j} [\,(u_j)_t + (F_j)_x\,]\ \phi\, dx = 0\,. \tag{7.25}$$

Since both u_j and $(F_j)_x$ are polynomials of degree $K-1$, and the above holds for any polynomial ϕ of degree $K-1$,

$$(u_j)_t + (F_j)_x = 0\,. \tag{7.26}$$

In conclusion, the DG scheme is identical to the flux reconstruction scheme with the Radau polynomial as correction functions. This completes the proof.

Note that the solution by DG method via (7.4) involves the inverse of the mass matrix due to the technique of projecting onto the basis functions $\phi_{j,k}$, which play the role of the test function. The solution via flux reconstruction approach using Eqs. (7.26) and (7.23), however, involves no mass matrix regardless of the choice of solution points since no test functions are employed.

Flux Reconstruction Schemes in DG Framework.

We now show the converse: for the case of advection, the flux reconstruction scheme can be cast in the DG framework. The difference, compared with standard DG schemes, is the space of test functions: for DG, this space is $\boldsymbol{P}_{K-1}$; for the scheme defined by the flux correction function $g = g_{\mathrm{LB}}$, in the case of advection equation $u_t + au_x = 0$ where $a > 0$, this space consists of polynomials ϕ of degree K such that $d\phi/dx$ is orthogonal to g.

To proceed, let g be defined by $g(-1) = 1$ and $g(1) = 0$, and g be a polynomial of degree K on $I = [-1,1]$ (approximating 0 in some sense, but we do not need this property). Applying (7.23) and (7.26) to the case of advection and using the local coordinate, we have

$$F_j(\xi) = f_j(\xi) + [\, f_{j-1/2,\mathrm{upw}}(-1) - f_j(-1)\,]\, g(\xi), \tag{7.27}$$

and

$$\frac{h_j}{2}(u_j)_t + (F_j)_\xi = 0. \tag{7.28}$$

To cast the above method in the DG framework, let ϕ be a polynomial of degree K on I. The above implies

$$\int_I [\, \frac{h_j}{2}(u_j)_t + (F_j)_\xi\,]\, \phi \; d\xi = 0. \tag{7.29}$$

Next, using (7.27),

$$\int_I (F_j)_\xi \, \phi \; d\xi \;=\; \int_I \left\{ f_j(\xi) + \left[f_{j-1/2,\mathrm{upw}}(-1) - f_j(-1) \right] g(\xi) \right\}_\xi \phi \;\; d\xi. \tag{7.30}$$

Separately, by integration by parts,

$$\int_I (f_j)_\xi \; \phi \; d\xi = (f_j\,\phi)(1) - (f_j\,\phi)(-1) - \int_I f_j(\xi)\, \phi_\xi \;\; d\xi, \tag{7.31}$$

Again, by integration by parts, since $g(-1) = 1$ and $g(1) = 0$,

$$\int_I g_\xi \phi \;\; d\xi = -\phi(-1) - \int_I g \; \phi_\xi \;\; d\xi. \tag{7.32}$$

Using the above two equations, (7.30) implies

$$\begin{aligned} \int_I (F_j)_\xi \, \phi \; d\xi \;=\; & (f_j\,\phi)(1) - (f_j\,\phi)(-1) - \int_I f_j(\xi)\, \phi_\xi \;\; d\xi \;-\; \left[f_{j-1/2,\mathrm{upw}}(-1) - f_j(-1) \right] \phi(-1) \\ & - \int_I \left[f_{j-1/2,\mathrm{upw}}(-1) - f_j(-1) \right] g(\xi)\, \phi_\xi \;\; d\xi \end{aligned} \tag{7.33}$$

For the moment, assume that ϕ satisfies

$$\int_I g \; \phi_\xi \;\; d\xi = 0. \tag{7.34}$$

Observing that $f_{j+1/2,\mathrm{upw}} = f_j(1)$ since the wind blows from the left, (7.33) then implies,

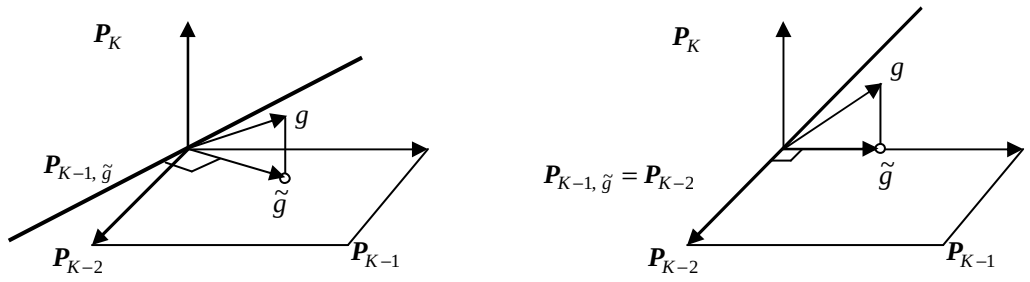
$$\int_I (F_j)_\xi \, \phi \; d\xi \;=\; f_{j+1/2,\mathrm{upw}}(1)\,\phi(1) - \; f_{j-1/2,\mathrm{upw}}(-1)\,\phi(-1) - \int_I f_j \, \phi_\xi \; d\xi. \tag{7.35}$$

Equation (7.29) and the above yield the following DG form for the solution:

$$\frac{\partial}{\partial t} \int_I \frac{h_j}{2} u_j \, \phi \, d\xi + \left[f_{\mathrm{upw}} \, \phi \right]_{\partial I} - \int_I f_j \, \phi_\xi \, d\xi = 0. \tag{7.36}$$

This equation is identical to (7.4) except for the space of test functions denoted by $\boldsymbol{T}$. This space consists of polynomials ϕ of degree K such that, by (7.34), $\phi' = \phi_\xi$ is orthogonal to g.

To describe $\boldsymbol{T}$, first we show that it is a subspace. Indeed, if ϕ and φ are in $\boldsymbol{T}$, then it can easily be verified that $\alpha\phi + \beta\varphi$ is also in $\boldsymbol{T}$ for any real number α and β.



(a) $P_{K-1,\tilde{g}}$ has the same dimension as P_{K-2} (b) The case g is the Radau polynomial

Figure 7.1. (a) $P_{K-1,\tilde{g}}$ **is of dimension** $K-1$**.** *Here,* $\tilde{g}$ *is the projection of* g *onto* P_{K-1}, and $P_{K-1,\tilde{g}}$ *is the subspace consisting of all* φ *in* P_{K-1} *that are orthogonal to* $\tilde{g}$. **(b)** *If* $g = R_{R,K}$, *then* $P_{K-1,\tilde{g}} = P_{K-2}$*; therefore,* $T = P_{K-1}$.

Next, we show that T is of dimension K. To this end, recall that P_K is of dimension $K+1$, and P_{K-1}, dimension K. Denote the projection of g onto P_{K-1} by $\tilde{g}$. On P_{K-1}, the collection of all φ that are orthogonal to $\tilde{g}$ form a subspace denoted by $P_{K-1,\tilde{g}}$, which is of dimension $K-1$. Since ϕ' is in P_{K-1} and g is in P_K, the condition that ϕ' is orthogonal to g is equivalent to the condition that ϕ' is orthogonal to $\tilde{g}$, i.e., ϕ' lies in $P_{K-1,\tilde{g}}$. By observing that the derivative operator is a linear mapping from P_K onto P_{K-1}, which maps T onto $P_{K-1,\tilde{g}}$, and $P_{K-1,\tilde{g}}$ is of dimension $K-1$, we conclude that T is of dimension K. This completes the proof.

VIII. Two Dimensional Extensions

For the two-dimensional (2D) extension of the current flux reconstruction approach, the solution procedure reduces to a series of one-dimensional (1D) operations in a manner similar to the extension of the staggered-grid scheme discussed in (Kopriva and Kolias 1996). Here, however, the schemes are simpler since they involve only one grid instead of three.

With $u = u(x, y, t)$, the 2D conservation law takes the form

$$u_t + f_x + g_y = 0 \tag{8.1}$$

with initial condition

$$u(x, y, 0) = u_{\text{init}}(x, y). \tag{8.2}$$

We assume that u is periodic or of compact support so that boundary conditions are trivial. Denote

$$\vec{f} = (f, g) \quad \text{and} \quad \text{div}(\vec{f}) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right).(f, g) = f_x + g_y. \tag{8.3a, b}$$

Then the conservation law can be written as

$$u_t + \text{div}(\vec{f}) = 0. \tag{8.4}$$

The simplest nontrivial two-dimensional conservation law is the advection equation: with constants a and b,

$$u_t + a\,u_x + b\,u_y = 0. \tag{8.5}$$

We will discuss the general case of unstructured quadrilateral meshes later. For the moment, let the domain of calculation be divided into uniform cells; each cell is a rectangle $E_{i,j}$ of length h_x, width h_y, and center $(x_{i,j}, y_{i,j})$, $i, j = \ldots -1, 0, 1, 2, \ldots$ Next, in the local description, let the solution points on $I = [-1, 1]$ be ξ_k, $k = 1, \ldots, K$. For simplicity, we use the same solution points in the η direction: $\eta_l = \xi_l$, $l = 1, \ldots, K$. The $K \times K$ solution points on the biunit square $I \times I = I^2$ are given by the tensor product of the 1D solution points, namely, (ξ_k, η_l) where $k = 1, \ldots, K$ and $l = 1, \ldots, K$. The solution points on $E_{i,j}$ are defined by

$$(x_{i,j,k,l}, y_{i,j,k,l}) = (x_{i,j} + \xi_k\, h_x/2,\ y_{i,j} + \eta_l\, h_y/2).$$

On each rectangle $E_{i,j}$, let the solution u be approximated by $K \times K$ pieces of data $u_{i,j,k,l}$ at the solution points $(x_{i,j,k,l}, y_{i,j,k,l})$, $k, l = 1,...,K$.

At time level *n*, suppose the data $u_{i,j,k,l}$ are known for all *i*, *j*, *k*, and *l*. We wish to calculate $d\,u_{i,j,k,l}(t)/dt$ at time $t = t^n$, which is abbreviated to $d\,u_{i,j,k,l}/dt$, in terms of the data. In other words, we wish to evaluate f_x and g_y at the solution points. Then, we march in time by, say, a Runge-Kutta method.

First, on $E_{i,j}$ we construct a polynomial interpolating $u_{i,j,k,l}$ where k and l vary. Equivalently, in the local coordinates (ξ,η) on I^2, we wish to construct a polynomial interpolating the data $u_{i,j,k,l}$ at the solution points (ξ_k,η_l) where k and l vary. To this end, with ϕ_k denoting the basis function for the 1D case given in (2.7b), let the basis functions on the biunit square I^2 be defined via tensor product: for each fixed (k,l), $k, l = 1,...,K$,

$$\phi_{k,l} = \phi_{k,l}(\xi,\eta) = \phi_k(\xi)\phi_l(\eta).$$

Thus, $\phi_{k,l}$ takes value 1 at the solution point (ξ_k,η_l) and 0 at all other K^2-1 solution points. See Fig. 8.1. The polynomial interpolating the data on the cell $E_{i,j}$ is

$$u_{i,j}(\xi,\eta) = \sum_{k,l=1}^{K} u_{i,j,k,l}\,\phi_{k,l}(\xi,\eta) = \sum_{k,l=1}^{K} u_{i,j,k,l}\,\phi_k(\xi)\phi_l(\eta). \tag{8.6}$$

Note that for each fixed η, $u_{i,j}(\xi,\eta)$ is a polynomial of degree $K-1$ in ξ, and vice versa. That is, $u_{i,j}$ is in $\boldsymbol{P}_{K-1,K-1} = \boldsymbol{P}_{K-1} \otimes \boldsymbol{P}_{K-1}$; e.g., for $K = 2$, $u_{i,j}$ is bilinear; for $K = 4$, bicubic.

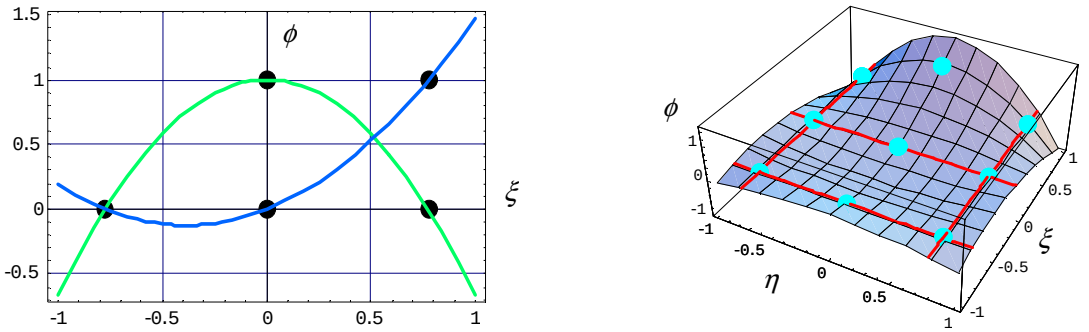


Figure 8.1. Basis functions for $K = 3$ with Gauss points as solution points. (a) The functions ϕ_2 and ϕ_3 for the 1D case. (b) The function $\phi_{3,2} = \phi_3(\xi)\phi_2(\eta)$ for the 2D case.

Let the solution point in the η direction be fixed, say $l = 2$, i.e., $\eta = \eta_2$. See Fig. 8.2(a). Along the ξ direction, the K values $u_{k,2}$, where $1 \le k \le K$, determine a polynomial of degree $K-1$ in ξ. Noting that $\phi_2(\eta_2) = 1$ and $\phi_l(\eta_2) = 0$ for $l \ne 2$, the above equation yields

$$u_{i,j}(\xi,\eta_2) = \sum_{k,l=1}^{K} u_{i,j,k,l}\,\phi_k(\xi)\phi_l(\eta_2) = \sum_{k=1}^{K} u_{i,j,k,2}\,\phi_k(\xi). \tag{8.7}$$

Thus, for each fixed η_l, the interpolation along the ξ-direction reduces to a one-dimensional problem. (A similar statement holds for each fixed ξ_k, along the η-direction.) We can calculate the left and right cell edge values $u_{i,j}(-1,\eta_2)$ and $u_{i,j}(1,\eta_2)$ by the 1D procedure. Along each vertical edge of midpoint $(i+1/2, j)$ (see Fig. 8.2(a)), using the data in the adjacent cells, i.e., in $E_{i-1,j}$ and $E_{i,j}$, at each η_l, say $\eta_l = \eta_2$, we can obtain the values just to its left and right:

$$u_L = u_{i,j}(1, \eta_2) \quad \text{and} \quad u_R = u_{i+1,j}(-1, \eta_2)\,. \tag{8.8}$$

From these two values, we can obtain the upwind flux normal to the edge:

$$f_{\text{upw}} = f_{\text{upw}}(u_L, u_R)\,. \tag{8.9}$$

Next, we define the discontinuous flux functions. At each solution point, from $u_{i,j,k,l}$, we can evaluate the corresponding fluxes:

$$f_{i,j,k,l} = f(u_{i,j,k,l}) \quad \text{and} \quad g_{i,j,k,l} = g(u_{i,j,k,l})\,. \tag{8.10}$$

On each cell $E_{i,j}$, the polynomial $f_{i,j}$ interpolating the values $f_{i,j,k,l}$ is given by

$$f_{i,j}(\xi,\eta) = \sum_{k,l=1}^{K} f_{i,j,k,l}\,\phi_{k,l}(\xi,\eta) = \sum_{k,l=1}^{K} f_{i,j,k,l}\,\phi_k(\xi)\phi_l(\eta)\,. \tag{8.11}$$

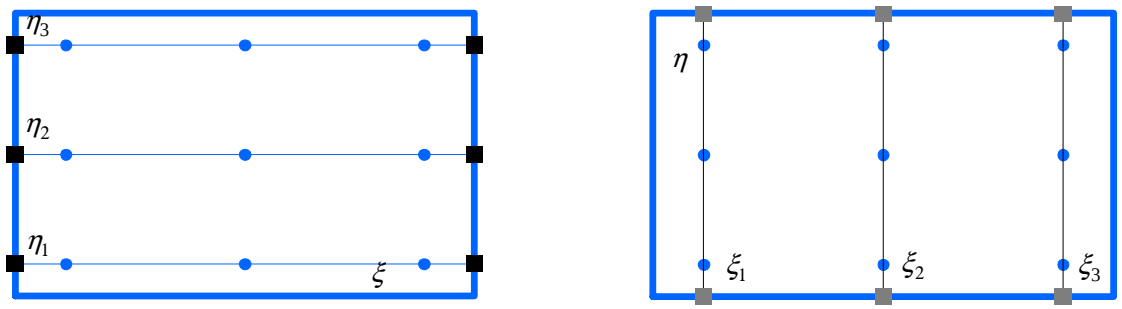
Similarly,

$$g_{i,j}(\xi,\eta) = \sum_{k,l=1}^{K} g_{i,j,k,l}\,\phi_{k,l}(\xi,\eta) = \sum_{k,l=1}^{K} g_{i,j,k,l}\,\phi_k(\xi)\phi_l(\eta)\,. \tag{8.12}$$

Again, let the solution point η_l be fixed, say, $l = 2$. Along the ξ direction, the K values $f_{k,2}$ (where k varies) determine a polynomial of degree $K-1$ in ξ

$$f_{i,j}(\xi, \eta_2) = \sum_{k=1}^{K} f_{i,j,k,2}\,\phi_k(\xi)\,. \tag{8.13}$$

We can calculate the flux values at the left and right interfaces by $f_{i,j}(-1, \eta_2)$ and $f_{i,j}(1, \eta_2)$.



(a) (b)

Figure 8.2. (a) Solution points (dots) and flux points on vertical boundaries (squares). *At each of these flux points, we define an upwind flux value* f_{upw} **. (b) Flux points on horizontal boundaries (squares)**. *Here, we define* g_{upw}.

Next, to evaluate f_x or equivalently (up to a constant factor) f_ξ for each fixed η_l, say, η_2, we construct a polynomial $F_{i,j}(\xi, \eta_2)$ for the flux, which is of degree K in ξ and take the upwind flux values f_{upw} at the two interfaces corresponding to $\xi = -1$ and $\xi = 1$. The procedure is exactly the same as that in Section 3. The calculation of g_y is similar; we omit the details.

The solution procedure can be summarized as follows. First, calculate and store the K upwind (normal) fluxes at each edge for all edges. Next, focusing on the cell $E_{i,j}$, evaluate and store $(f_\xi)_{i,j,k,l}$ and $(g_\eta)_{i,j,k,l}$ of the discontinuous flux functions for all k and l. To correct for the left and right edges, for each fixed l, add the correction terms to $(f_\xi)_{i,j,k,l}$ for $k = 1,\ldots,K$ as in the 1D case. In a similar manner, to correct for the bottom and top edges, for each fixed k, add the correction terms to $(g_\eta)_{i,j,k,l}$ for $l = 1,\ldots,K$. This completes the algorithm.

For the general case, we map the biunit square to a quadrilateral (with possibly curved edges) via an isoparametric mapping. The problem then reduces to one similar to the above rectangular mesh case. The details are again omitted.

Next, we carry out the Fourier (Von Neumann) analysis for the 2D case. Consider the advection equation

$$u_t + a\,u_x + b\,u_y = 0\,. \tag{8.14}$$

where $a = \cos(\alpha)$, $b = \sin(\alpha)$, and α is the flow angle. Note that the flow speed is 1. The cells employed for this analysis are squares centered at (i, j): $E_{i,j} = [i-1/2, i+1/2]\times[j-1/2, j+1/2]$. Due to symmetry, we only need to consider α in $[0, \pi/4]$. Thus, for the vertical edge identified by its midpoint $(i+1/2, j)$ and the horizontal edge $(i, j+1/2)$, the upwind values are the values from the cell $E_{i,j}$. Denote by $\boldsymbol{u}_{i,j}$ the column vector of K^2 components

$$\boldsymbol{u}_{i,j} = (u_{i,j,1,1}, u_{i,j,1,2}, ..., u_{i,j,1,K}, u_{i,j,2,1}, ..., u_{i,j,2,K}, ..., u_{i,j,K,1}, ..., u_{i,j,K,K})\,. \tag{8.15}$$

Similar to the 1D case, the solution can be expressed as

$$\frac{d\boldsymbol{u}_{i,j}}{dt} = \boldsymbol{C}_{0,0}\,\boldsymbol{u}_{i,j} + \boldsymbol{C}_{-1,0}\,\boldsymbol{u}_{i-1,j} + \boldsymbol{C}_{0,-1}\,\boldsymbol{u}_{i,j-1}\,. \tag{8.16}$$

Here, $\boldsymbol{C}_{0,0}$, $\boldsymbol{C}_{-1,0}$, and $\boldsymbol{C}_{0,-1}$ are $K^2 \times K^2$ matrices. Let the wave number in the x-direction be w_x and that in the y-direction, w_y. Let the imaginary number be denoted by $\boldsymbol{I}$ instead of the standard notation i to avoid confusion with the index i. Replacing $\boldsymbol{u}_{i-1,j}$ by $e^{-\boldsymbol{I}\,w_x}\boldsymbol{u}_{i,j}$ and $\boldsymbol{u}_{i,j-1}$ by $e^{-\boldsymbol{I}\,w_y}\boldsymbol{u}_{i,j}$, we obtain

$$\frac{d\boldsymbol{u}_{i,j}}{dt} = \left(\boldsymbol{C}_{0,0} + e^{-\boldsymbol{I}\,w_x}\boldsymbol{C}_{-1,0} + e^{-\boldsymbol{I}\,w_y}\boldsymbol{C}_{0,-1}\right)\boldsymbol{u}_{i,j}\,. \tag{8.17}$$

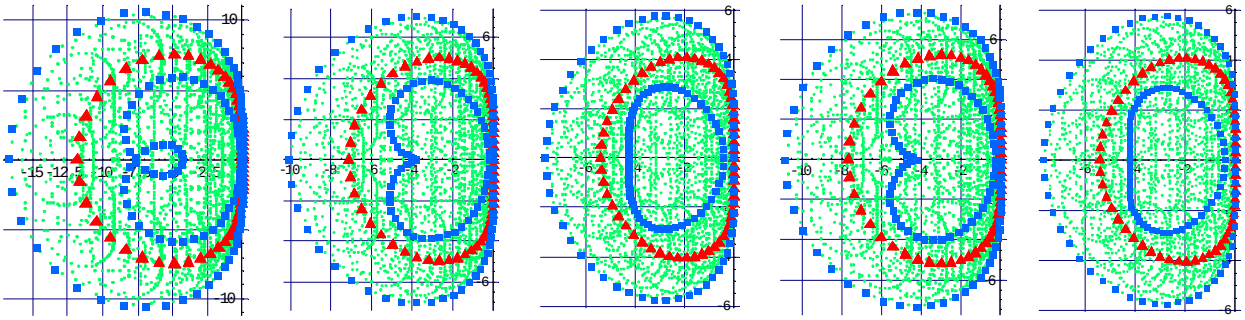
Therefore, the semidiscretization results in the equation

$$\frac{d\boldsymbol{u}_{i,j}}{dt} = \boldsymbol{S}\,\boldsymbol{u}_{i,j} \tag{8.18}$$

For all schemes discussed here, $\boldsymbol{S}$ has K^2 eigenvalues. The eigenvalue which approximates the exact one is the *principal eigenvalue* denoted by $S(w_x, w_y)$:

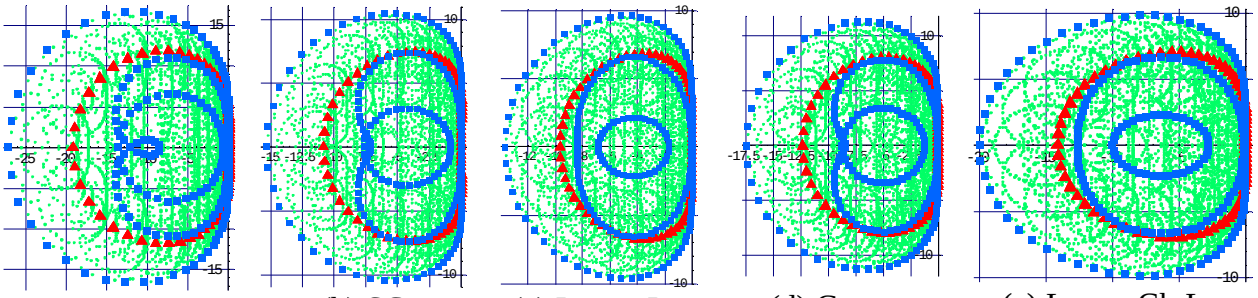
$$S(w_x, w_y) \approx -i(aw_x + bw_y)\,. \tag{8.19}$$

All other eigenvalues are spurious ones. All eigenvalues must lie in the left half of the complex plane for the semidiscretization to be stable.



(a) DG (b) SG (c) Lump, Lo (d) Gauss (e) Lump, Ch-Lo

Figure 8.3. Spectra (collection of all eigenvalues) of 2D semidiscrete schemes, $K = 3$**.** *Note that if the flow is horizontal, then the solution reduces to that of the 1D case, and so does the spectrum. The spectrum for the case of horizontal flow is shown by the small triangles; diagonal flow, small squares; and all flow directions, small dots. For all tested cases, the boundary of the whole spectrum is* $\sqrt{2}$ *times the spectrum of the 1D case (except when the 1D spectrum is not convex as in the case of* $g_{\text{Lump, Ch-Lo}}$ *with* $K = 8$*). The spectrum for the case of diagonal flow also yields the boundary of the whole spectrum.*



(a) DG (b) SG (c) Lump, Lo (d) Gauss (e) Lump, Ch-Lo

Figure 8.4. Spectra of 2D semidiscrete schemes, $K = 4$.

The spectra (collection of all eigenvalues) of the semidiscrete schemes are shown in Fig. 8.3 for the case $K = 3$, and in Fig. 8.4, $K = 4$. Note that if the flow is horizontal, then the solution reduces to that of the 1D case, and so does the spectrum. The spectrum for the case of horizontal flow is shown by the small triangles; diagonal flow, small squares; and all flow directions, small dots. For all tested cases, the boundary of the whole spectrum is $\sqrt{2}$ times the spectrum of the 1D case (except when the 1D spectrum is not convex as in the case of $g_{\text{Lump, Ch-Lo}}$ with $K = 8$). The spectrum for the case of diagonal flow also yields the boundary of the whole spectrum.

Next, we discuss accuracy. The orders of accuracy and errors of the 2D schemes for $K = 4$ are tabulated in Table 8.1. Note that the order of accuracy for the 2D case is identical to that of the 1D case for all schemes. The errors for schemes DG, g_{Ga}, $g_{\text{Lump, Lo}}$, and $g_{\text{Lump, Ch-Lo}}$ (lumping for Chebyshev-Lobatto points), have negative real parts; a closer check of the eigenvalues reveals that these four schemes are stable. The errors for the last three schemes, namely, SG, g_3, and g_{Lo} have positive real parts; as a result, they are unstable, but the instability is mild.

($K = 4$) Scheme in 2D	Order of accuracy	Coarse mesh error, $\alpha = \pi/7$ $w_x = \pi/3,\ w_y = \pi/5$	Fine mesh error, $\alpha = \pi/7$ $w_x = \pi/6,\ w_y = \pi/10$
DG	7	$-8.9\times10^{-7} - 1.2\times10^{-7}\,i$	$-3.6\times10^{-9} - 2.4\times10^{-10}\,i$
g_{Ga}	6	$-2.7\times10^{-6} + 8.7\times10^{-6}\,i$	$-1.1\times10^{-8} + 7.2\times10^{-8}\,i$
$g_2 = g_{\text{Lump, Lo}}$	6	$-4.8\times10^{-6} + 1.5\times10^{-5}\,i$	$-2.\times10^{-8} + 1.3\times10^{-7}\,i$
$g_{\text{Lump, Ch-Lo}}$	4	$-1.7\times10^{-4} + 3.7\times10^{-4}\,i$	$-2.7\times10^{-6} + 1.2\times10^{-5}\,i$
SG	4	$5.9\times10^{-5} - 7.6\times10^{-5}\,i$	$1.\times10^{-6} - 3.0\times10^{-6}\,i$
g_3	5	$2.8\times10^{-4} + 1.7\times10^{-4}\,i$	$5.0\times10^{-6} + 1.4\times10^{-6}\,i$
g_{Lo}	4	$1.0\times10^{-4} - 1.4\times10^{-4}\,i$	$1.7\times10^{-6} - 5.2\times10^{-6}\,i$

Table 8.1. Errors and accuracy of 2D schemes for flow angle $\alpha = \pi/7$ and $K = 4$.

IX. Numerical Results

As a preliminary test we apply the schemes discussed above to solve the advection equation. Periodic boundary conditions and the standard 4-stage Runge-Kutta time stepping are employed for all problems. (Recall that the correction function $g_{\text{Lump, Lo}}$ uses the Lobatto points as solution points and lumps the correction to the corresponding boundary point; similarly, $g_{\text{Lump, Ch-Lo}}$ uses the Chebyshev-Lobatto points and lumps the correction in the same manner; and the correction function g_{Ga} vanishes at the $K-1$ Gauss points.) The solutions by the schemes DG, SG, $g_{\text{Lump, Lo}}$, g_{Ga}, $g_{\text{Lump, Ch-Lo}}$ and, for comparison, the standard piecewise-linear upwind scheme (i.e., Van Leer's MUSCL or Fromm's scheme) are shown. For all plots, the crosses represent the exact solutions and the dots, the numerical solutions.

The three tests involve the advection of a Gaussian wave. On the domain $[0,1]$, the initial condition is

$$u_{\text{init}}(x) = \exp[-40(x-.5)^2]. \tag{9.1}$$

We employ 10 cells and, in each cell, four solution points. Thus, the wave is represented by a total of 40 data points.

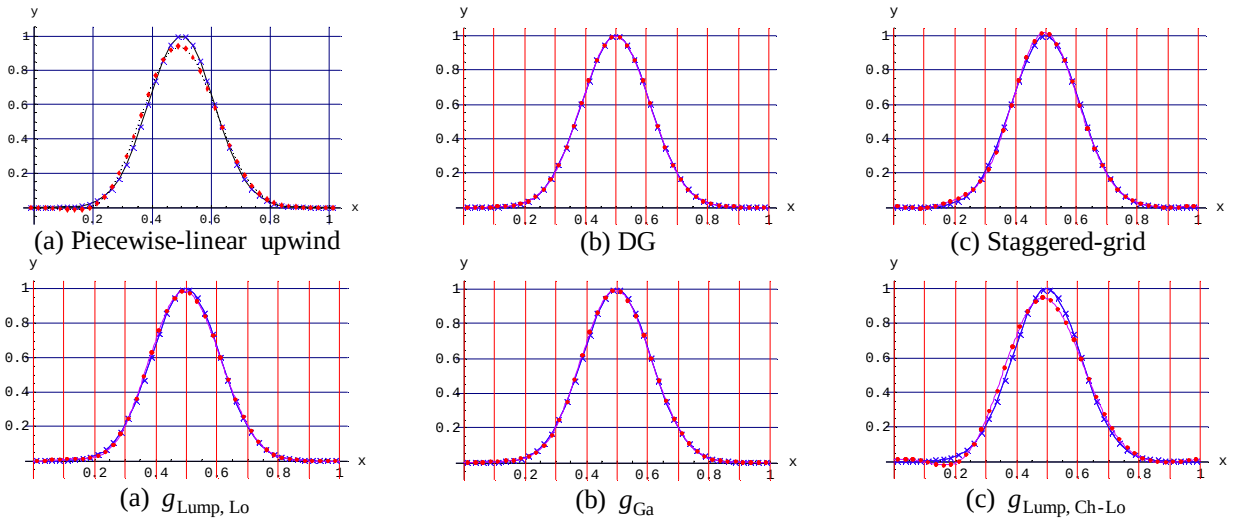


Figure 9.1. Advection of a Gaussian wave. *The wave travels a distance of 10 periods or 400 cells; the solution points are the equidistant points (not including the two cell boundaries); the crosses represent the exact solutions and the dots, the numerical solutions.*

For the first test, equidistant (not including the cell boundaries) points are used as solution points. The wave travels a distance of 10 periods, i.e., 400 cells. Each time step corresponds to a CFL number of approximately .9*(CFL limit) in a manner that the distance of 10 periods is an integer multiple of the time step size. The numbers of time steps are: for DG, 764; SG, 432; $g_{\text{Lump, Lo}}$, 385; g_{Ga}, 490; and $g_{\text{Lump, Ch-Lo}}$, 562. For the piecewise-linear upwind scheme, a CFL number of .750469 is employed; after 533 time steps, the wave traveled a distance of 400 cells. The solutions are shown in Fig. 9.1. Note that the DG scheme yields the most accurate solution; on the other hand, the piecewise-linear upwind and $g_{\text{Lump, Ch-Lo}}$ schemes yield the least accurate ones.

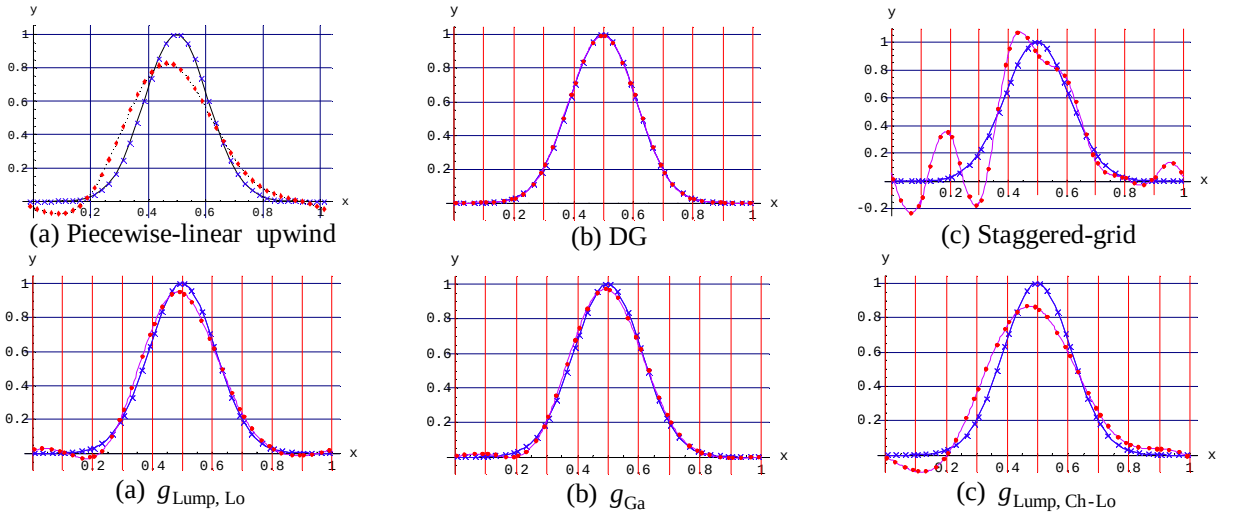


Figure 9.2. Advection of a Gaussian wave for a long distance. *The wave travels a distance of 50 periods or 2000 cells; the solution points are the Gauss points. The DG scheme yields the most accurate solution; on the other hand, the piecewise-linear upwind, staggered-grid, and $g_{\text{Lump, Ch-Lo}}$ schemes yield the least accurate ones. Note that the staggered-grid solution exhibits growing wiggles, which are a sign of instability which, in this case, is mild.*

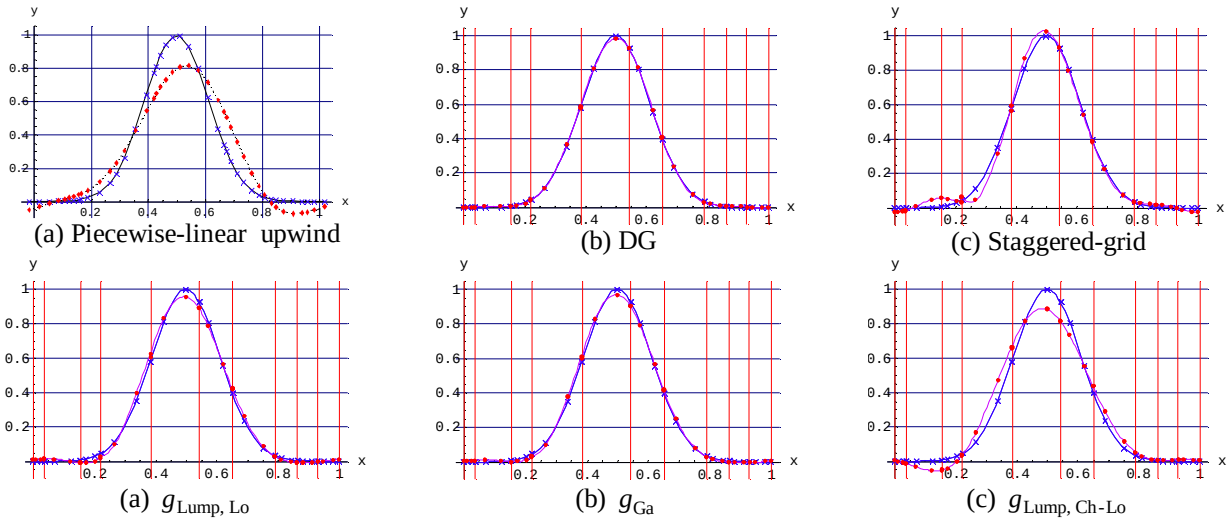


Figure 9.3. Advection of a Gaussian wave with an irregular mesh. *The wave travels a distance of 10 periods or 400 cells. The mesh is irregular (random). The solution points are the Lobatto points.*

The second test case is the same as the first except that the solution points are the Gauss points, and the wave travels a distance of 50 periods, which corresponds to 2000 cells. For the piecewise-linear upwind scheme, a uniform mesh is employed with a CFL number of .7499; after 2667 time steps, the wave travels a distance of 2000 cells. The solutions are shown in Fig. 9.2. Again, note that the most accurate solution is that of DG; among the least accurate solutions are those of the piecewise-linear upwind and $g_{\text{Lump, Ch-Lo}}$ schemes. Also note that the staggered-grid solution exhibits growing wiggles, which are a sign of (mild) instability. Here, the solution is identical to that of the original staggered-grid method.

The third test is also the same as the first except that the grids are generated randomly with the constraint that the smallest cell is at least 1/5 the size of the largest. Here, except for the piecewise-linear upwind scheme, we use the Lobatto points as solution points for all schemes. Note that the solutions are much less accurate than those of the first test, which employ a uniform mesh. In general, the solutions are comparable to those in Fig. 9.2 where the wave travels a distance five times longer.

X. Conclusions and Discussion

In summary, we introduced a new approach to high-order accuracy for the numerical solution of conservation laws via the construction of a continuous flux function. This function is obtained by adding correction terms, which account for the jumps in flux values at interfaces, to the discontinuous flux function. The approach bridges the two existing approaches of DG and staggered-grid (or spectral difference) and leads to simplified versions of these two methods. Schemes via the new approach are simple and economical: they solve the conservation laws in differential form and involve only one grid, namely, the solution points; they are also conservative.

Among the new schemes (with K solution points), two are super-accurate of order $2K-2$, one order less than that of DG. The first is the scheme with the correction function $g_{\text{Lump, Lo}}$ that employs the Lobatto points as solution points and lumps the correction to the corresponding boundary point. The second is the scheme with the correction function g_{Ga}, which vanishes at the $K-1$ Gauss points. Compared to the DG scheme, these two schemes have the advantage that the corresponding CFL limits are not as restrictive: for $g_{\text{Lump, Lo}}$, it is roughly twice as large, and for g_{Ga}, about three halves.

Concerning the property of super-accuracy or super-convergence, the following result was obtained via Fourier (Von Neumann) analysis. With K solution points, if the correction function g is perpendicular to $\boldsymbol{P}_m$ where $0 \le m \le K-2$, then the scheme is accurate to order $K+m+1$.

The relation between the DG and new approaches was established: (1) the DG method can be cast via the new approach, and (2) for an advection problem, schemes via the new approach can be cast in the DG framework. As a result of the new approach, the DG scheme can be formulated using the differential instead of integral form.

Next, we discuss extensions and applications.

1. The approach can be extended to the case of the Euler and Navier-Stokes equations on a 2D quadrilateral mesh in a manner similar to the extension of the staggered-grid schemes (Kopriva 1998). The extension to the case of a triangular mesh, however, remains to be explored. The key problem in the triangular mesh case is that the concept of tensor product is not available.

2. The approach can be applied to solve ordinary differential equations (ODE). If we use the right Radau points as solution points in the time direction and the right Radau polynomial as the correction function, the result is an implicit Runge-Kutta scheme identical to the Radau IIA scheme (Hairer and Wanner 1991). The proof using our approach is simpler than that of LeSaint and Raviart (1974) where a quadrature formula including the left boundary was employed. Our approach is more direct: it employs a formula including the right boundary.

3. Using the application to ODE discussed above and some additional techniques introduced for the moment scheme in (Huynh 2004) and employed in (Suzuki and Van Leer, 2007), one can derive a moment scheme of degree K for arbitrary K, which is explicit and has a CFL limit of 1 for the case of one spatial dimension. Such a scheme can be considered as an extension of Van Leer's scheme 3 (1977), which is a piecewise linear scheme for advection. It can also be considered as an explicit space-time DG scheme with the property that the CFL limit is 1, whereas the CFL limit for the combination of a method of line (or Runge-Kutta) and DG is considerably smaller.

4. Concerning the recent spectral difference and spectral volume schemes on a triangular mesh, two questions can be raised. (1) Are these schemes stable or mildly unstable? (2) Can we derive spectral schemes that share the properties of stability and super-accuracy of the scheme g_{Ga} discussed above?

Appendix: Polynomials, Quadratures, and Related Topics

Since the Legendre, Radau, and Lobatto polynomials play a key role in our approach, they are reviewed below. We also introduce new polynomials and derive additional properties needed.

A. Legendre Polynomials

For any smooth functions v and w on the interval $I=[-1,1]$, let the L^2 inner product be defined by

$$(v,w)=\int_{-1}^{1} v(\xi)w(\xi)d\xi\,. \tag{A.1}$$

Let m be a nonnegative integer, and let $\boldsymbol{P}_m$ be the space of polynomials of degree m or less. Then $\boldsymbol{P}_m$ is a vector space of dimension $m+1$. The $m+1$ monomials $1,\xi,\ldots,\xi^m$ form a basis for $\boldsymbol{P}_m$, which is not orthogonal. Using the Gram-Schmidt process, we can derive an orthogonal basis from a nonorthogonal one. The result is the Legrendre polynomials P_k, $k=0,1,2,\ldots$ where P_k is scaled so that $P_k(1)=1$. The Legendre polynomial of degree k is given by the following recurrence formula (see, e.g., Szego 1967 or Hildebrand 1974):

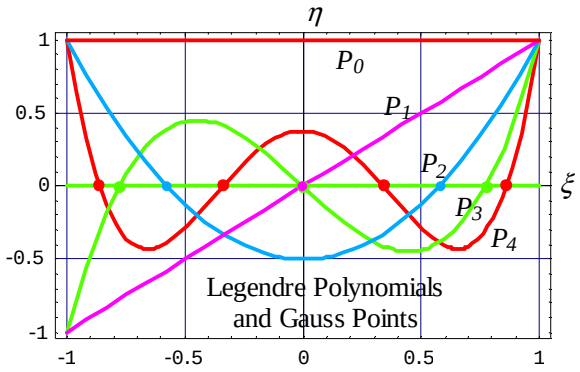


Figure A.1. Legendre Polynomials and Gauss points. *Note the different scales in ξ and η axes.*

$$P_{-1}=0,\quad P_0=1,$$

and, for $k\ge 1$,

$$P_k(\xi)=\frac{2k-1}{k}\,\xi\,P_{k-1}(\xi)-\frac{k-1}{k}\,P_{k-2}(\xi)\,. \tag{A.2}$$

From their definitions, the Legendre polynomials satisfy, for $k\ne l$,

$$(P_k,P_l)=\int_{-1}^{1} P_k(\xi)P_l(\xi)d\xi=0\,. \tag{A.3}$$

For $l=k$, we have the following property, which is useful in the standard formulation of DG scheme, but is not employed here,

$$\gamma_k=(P_k,P_k)=\int_{-1}^{1}(P_k(\xi))^2 d\xi=\frac{2}{2k+1}\,. \tag{A.4}$$

The first few Legendre polynomials are

$$P_0 = 1, \quad P_1 = \xi, \quad P_2 = \tfrac{1}{2}(3\xi^2 - 1), \quad P_3 = \tfrac{1}{2}(5\xi^3 - 3\xi), \quad P_4 = \tfrac{1}{8}(35\xi^4 - 30\xi^2 + 3) \ .$$

The plots of these polynomials are shown in Fig. A.1. Note that $P_k(1) = 1$ for all k, while $P_k(-1) = 1$ for even k, and $P_k(-1) = -1$ for odd k; in addition, P_k is an even function (involves only even powers of ξ) for even k, and an odd function for odd k.

Next, for nonnegative integers k and m, if $k > m$, then the Legendre polynomial P_k is perpendicular to $\boldsymbol{P}_m$, the space of polynomials of degree m or less. In particular, P_{m+1} is perpendicular to $\boldsymbol{P}_m$, i.e., for any polynomial p of degree m or less,

$$(P_{m+1}, p) = \int_{-1}^{1} P_{m+1}(\xi)\, p(\xi)\, d\xi = 0 \ . \tag{A.5}$$

Readers who are not interested in the proofs can skip the rest of this subsection with no loss of continuity.

We will need the values $P_k'(-1)$ and $P_k'(1)$, which can be calculated as follows. First, rewriting Eq. (A.2) with k replaced by $k+1$, we have

$$P_{k+1}(\xi) = \frac{2k+1}{k+1}\, \xi\, P_k(\xi) - \frac{k}{k+1}\, P_{k-1}(\xi) \ . \tag{A.6}$$

Differentiating the above,

$$P_{k+1}'(\xi) \ = \ \frac{2k+1}{k+1}\, [\, P_k(\xi) + \xi\, P_k'(\xi)\,] \ - \ \frac{k}{k+1}\, P_{k-1}'(\xi) \ .$$

Evaluating at $\xi = 1$, we obtain

$$P_{k+1}'(1) \ = \ \frac{2k+1}{k+1} + \frac{2k+1}{k+1} P_k'(1) \ - \ \frac{k}{k+1}\, P_{k-1}'(1) \ .$$

Subtracting $P_k'(1)$ from both sides,

$$P_{k+1}'(1) - P_k'(1) \ = \ \frac{2k+1}{k+1} + \frac{k}{k+1} P_k'(1) - \frac{k}{k+1}\, P_{k-1}'(1) \ .$$

Multiplying by $k+1$,

$$(k+1)[\, P_{k+1}'(1) - P_k'(1)\,] \ = \ 2k+1 \ + \ k[\, P_k'(1) - P_{k-1}'(1)\,] \ .$$

Denoting $D_k = P_k'(1) - P_{k-1}'(1)$, the above implies

$$(k+1)\, D_{k+1} \ = \ k\, D_k + 2k + 1 \ .$$

As a result,

$$D_k \ = \ k \ .$$

By the definition of D_k, therefore,

$$P_k'(1) \ = \ P_{k-1}'(1) + k \ .$$

Since $P_0 = 1$, $P_0'(1) = 0$. The above implies, for $k > 0$,

$$P_k'(1) \ = \ 1 + 2 + \ldots + k \ = \ \tfrac{1}{2} k(k+1) \ . \tag{A.7}$$

As for the left boundary, since P_k is even for even k, and odd for odd k, $P_k'(-1) = (-1)^{k-1} P_k'(1)$, i.e.,

$$P_k'(-1) \ = \ \frac{(-1)^{k-1}}{2}\, k(k+1) \ . \tag{A.8}$$

Next, the following two relations (see (8.5.7) of Hildebrand 1974) between P_k' and P_k will be employed later (in deriving the key property of $g_{\text{Lump, Lo}}$): for $k \geq 1$,

$$(1 - \xi^2) P_k'(\xi) \ = \ -k\xi\, P_k(\xi) + k\, P_{k-1}(\xi) \ , \tag{A.9}$$

and

$$(1 - \xi^2) P_k'(\xi) \ = \ (k+1)\xi\, P_k(\xi) - (k+1)\, P_{k+1}(\xi) \ . \tag{A.10}$$

Consider the right hand sides of (A.9) and (A.10). The term $\xi\, P_k(\xi)$ drops out if we multiply (A.9) by $k+1$ and (A.10) by k, and then add the results,

$$(k+k+1)(1-\xi^2)P_k'(\xi) = -k(k+1)[P_{k+1}(\xi)-P_{k-1}(\xi)]. \tag{A.11}$$

That is, for $k \geq 1$,

$$P_{k+1}(\xi)-P_{k-1}(\xi) = -\frac{k+(k+1)}{k(k+1)}(1-\xi^2)P_k'(\xi). \tag{A.12}$$

B. Gauss Quadratures

Let $K \geq 1$ be an integer and f a smooth function on $I=[-1,1]$. The integral $\int_{-1}^{1} f(\xi)d\xi$ can be approximated using K evaluations of f by

$$\int_{-1}^{1} f(\xi)d\xi \approx \sum_{k=1}^{K} w_k f(\xi_k); \tag{A.13}$$

here, both the weights w_k and the sample points ξ_k are parameters, which are to be determined. Therefore, the total number of unknowns is $2K$. Setting $f=\xi^k$, $k=0,1,...,2K-1$, respectively, and requiring the above approximation to be exact results in a system of $2K$ equations with $2K$ unknowns. This system determines w_k and ξ_k, for $k=1,...,K$. The resulting formula $\sum_{k=1}^{K} w_k f(\xi_k)$ is called a Gauss quadrature. The values ξ_k, $k=1,...,K$ are the Gauss (quadrature) points.

With just K evaluations, the Gauss quadrature has a degree of precision of $2K-1$, i.e., for any polynomial p of degree $2K-1$ or less, the Gauss quadrature for p is exact:

$$\int_{-1}^{1} p(\xi)d\xi = \sum_{k=1}^{K} w_k p(\xi_k);$$

In general, a quadrature rule with K evaluations has a degree of precision no higher than $K-1$. For Gauss quadratures, however, with two Gauss points ($K=2$), the degree of precision is 3 (exact for cubics); with three Gauss points, the degree of precision is 5.

We are now ready to show that the Gauss points $\xi_1,...,\xi_K$ defined by the quadrature (A.13) are identical to the K zeros of the Legrendre polynomial P_K. Consider the polynomial q of degree K, which vanishes at the K Gauss points, and takes value 1 at $\xi=1$:

$$q(\xi) = \prod_{k=1}^{K} \frac{\xi-\xi_k}{1-\xi_k}. \tag{A.14}$$

We wish to prove that q is identical to P_K. Since the Legendre polynomial P_K is a polynomial of degree K determined by the conditions that it is orthogonal to $\boldsymbol{P}_{K-1}$ and $P_K(1)=1$, we need to show that q satisfies these conditions. By the definition of q above, we have $q(1)=1$. Next, let r be an arbitrary polynomial of degree $K-1$. Then qr is of degree $2K-1$ and vanishes at the K Gauss points. Since the Gauss quadrature has degree of precision $2K-1$, we conclude that $\int_{-1}^{1} qr\,d\xi = 0$. As a result, q is orthogonal to $\boldsymbol{P}_{K-1}$. This completes the proof.

For $K=1$, the Gauss point is 0, and for K up to 4, the Gauss points are

$K=2$ $\quad \xi_1=-0.577350 \quad \xi_2=0.577350=1/\sqrt{3}$

$K=3$ $\quad \xi_1=-0.774597 \quad \xi_2=0 \quad \xi_3=0.774597$

$K=4$ $\quad \xi_1=-0.861136 \quad \xi_2=-0.339981 \quad \xi_3=0.339981 \quad \xi_4=0.861136$

For a fixed K, the weights w_k's of the Gauss quadrature with K evaluations at the Gauss points ξ_k are given by

$$w_k = \frac{2}{(1-\xi_k{}^2)[P_K'(\xi_k)]^2} = \frac{2(1-\xi_k{}^2)}{(K+1)^2[P_{K+1}(\xi_k)]^2}. \tag{A.15}$$

For the case of four Gauss points, the weights are

$w_1 = 0.347855$ $w_2 = 0.652145$ $w_3 = 0.652145$ $w_4 = 0.347855$.

C. Radau Quadratures and Polynomials

Again, let $K \geq 1$ be an integer and f a smooth function on $I = [-1,1]$. Consider a quadrature formula that includes one of the two boundary points, say, the right boundary $\xi = 1$:

$$\int_{-1}^{1} f(\xi) d\xi \approx \sum_{k=1}^{K} w_k f(\xi_k) \quad \text{where} \quad \xi_K = 1, \tag{A.16}$$

All $K-1$ other sample points ξ_k are parameters (unknowns); all w_k, $k = 1,...,K$ are also parameters. The total number of parameters is therefore $2K-1$. To solve for these unknowns, we need $2K-1$ equations, which can be obtained by setting $f = \xi^k$, $k = 0,1,...,2K-2$, respectively, in the above. This system of equations yields all w_k's and $K-1$ values ξ_k, $k = 1,...,K-1$. These ξ_k together with $\xi_K = 1$ are called the right Radau points; the corresponding quadrature, the right Radau quadrature. The left Radau points and quadrature are defined in the same manner with the right boundary point replaced by left. Clearly, the left Radau points are reflections (mirror images) of the right Radau points about 0.

With K evaluations (including that at $\xi_K = 1$), the Radau quadrature has a degree of precision of $2K-2$ (as opposed to $2K-1$ of the Gauss quadrature), i.e., it is exact for any polynomial of degree $2K-2$ or less.

While the Gauss points are the zeros of the Legendre polynomials, the Radau points are the zeros of the Radau polynomials discussed below. First, for any fixed positive integer m, denote by Pr_m the projection from the space of polynomials onto $\boldsymbol{P}_m$, i.e., for any polynomial p, the polynomial $\mathrm{Pr}_m(p)$ is the least-squares approximation for p on I by a polynomial of degree m. Clearly, if p is of degree m or less, then $\mathrm{Pr}_m(p) = p$.

We can now define the Radau polynomials. To avoid the factor $(-1)^K$, we deal with the left Radau polynomials. Consider the Legendre polynomials P_K and P_{K-1}. By definition, they are perpendicular to $\boldsymbol{P}_{K-2}$, i.e., $\mathrm{Pr}_{K-2}(P_K) = \mathrm{Pr}_{K-2}(P_{K-1}) = 0$. Next, let p given by

$$p = \alpha P_K + \beta P_{K-1}$$

where α and β are parameters to be determined. Clearly, p is perpendicular to $\boldsymbol{P}_{K-2}$. If we require p to satisfy

$$p(-1) = 0 \quad \text{and} \quad p(1) = 1,$$

then, α and β are determined. Indeed, since $P_k(-1) = 1$ for even k, and $P_k(-1) = -1$ for odd k (see Fig. A.1), while $P_k(1) = 1$ for all k, we have $\alpha = \beta = 1/2$. The resulting p is the left Radau polynomial of degree K denoted by $R_{L,K}$:

$$R_{L,K} = \tfrac{1}{2}(P_K + P_{K-1}) , \tag{A.17}$$

$$R_{L,K}(-1) = 0 \quad \text{and} \quad R_{L,K}(1) = 1. \tag{A.18}$$

Here the letter R stands for 'Radau' and subscript L for 'left'. Thus, $R_{L,K}$ is a polynomial of degree K determined uniquely by the two conditions of (A.18) and $K-1$ conditions that it is perpendicular to $\boldsymbol{P}_{K-2}$.

We can now show that the left Radau points ξ_k, $k = 1,...,K$ (where $\xi_1 = -1$) defined by a quadrature formula similar to (A.16) are identical to the K zeros of $R_{L,K}$. Indeed, consider the polynomial q of degree K, which vanishes at the K left Radau points, and takes value 1 at the right boundary $\xi = 1$: $q(\xi)$ is given by expression (A.14) with the Gauss points replaced by the Radau points. We wish to show that q is identical to $R_{L,K}$. Since $R_{L,K}$ is uniquely determined by the conditions that it is orthogonal to $\boldsymbol{P}_{K-2}$ and $R_{L,K}(-1) = 0$ and $R_{L,K}(1) = 1$, it suffices to show that q satisfies these conditions. By its definition, q satisfies that latter two conditions. Next, let r be an arbitrary polynomial of degree $K-2$. Then qr is of degree $2K-2$ and vanishes at the Radau points. Therefore, due to the degree of precision of $2K-2$ of the Radau quadrature, we have $\int_{-1}^{1} qr\, d\xi = 0$. That is, q is orthogonal to $\boldsymbol{P}_{K-2}$. This completes the proof.

For a fixed K, the weights w_k's of the left Radau quadrature with K evaluations at the left Radau points ξ_k are given by, for $k = 1, \ldots, K$,

$$w_k = \begin{cases} \dfrac{1}{K^2}\,\dfrac{1-\xi_k}{[P_{K-1}(\xi_k)]^2} = \dfrac{1}{1-\xi_k}\,\dfrac{2}{[P_{K-1}'(\xi_k)]^2} & \text{if} \quad k \neq 1, \\ \dfrac{2}{K^2} & \text{if} \quad k = 1. \end{cases} \tag{A.19}$$

As for the right Radau polynomials of degree K denoted by $R_{R,K}$, which vanishes at the right boundary,

$$R_{R,K} = \frac{(-1)^K}{2}(P_K - P_{K-1}), \tag{A.20}$$

$$R_{R,K}(-1) = 1 \quad \text{and} \quad R_{R,K}(1) = 0. \tag{A.21}$$

For a fixed K, the weights w_k's of the right Radau quadrature with K evaluations at ξ_k are given by

$$w_k = \begin{cases} \dfrac{1}{K^2}\,\dfrac{1+\xi_k}{[P_{K-1}(\xi_k)]^2} = \dfrac{1}{1+\xi_k}\,\dfrac{2}{[P_{K-1}'(\xi_k)]^2} & \text{if} \quad k \neq K, \\ \dfrac{2}{K^2} & \text{if} \quad k = K. \end{cases} \tag{A.22}$$

We will also need the slopes of the Radau polynomials at the boundary points. Since for any integer $k > 0$, $P_k'(-1) = (-1)^{k-1}\sum_{l=1}^{k} l$, by expression (A.17) for $R_{L,K}$,

$$R_{L,K}'(-1) = \frac{1}{2}[P_K'(-1) + P_{K-1}'(-1)] = \frac{1}{2}\left[(-1)^{K-1}\sum_{l=1}^{K} l + (-1)^{K-2}\sum_{l=1}^{K-1} l\right].$$

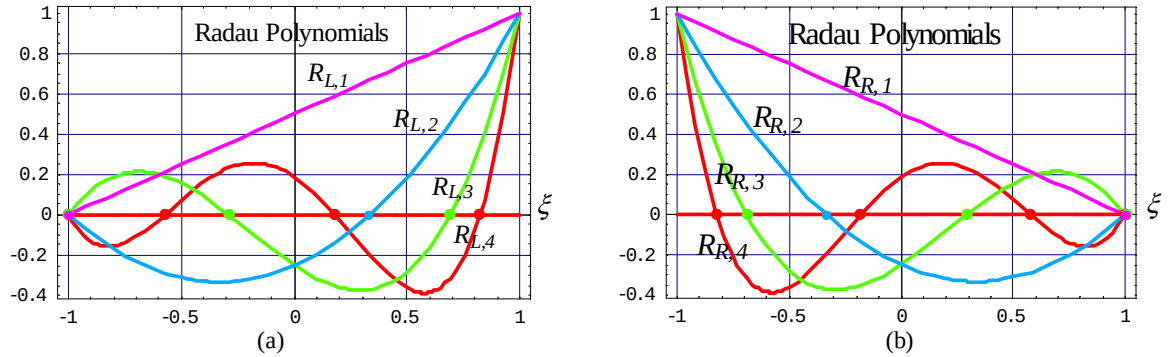


Figure A.2. Radau Polynomials on $I = [-1,1]$. (a) *Left;* (b) *Right.*

Noting that $(-1)^{K-1}$ and $(-1)^{K-2}$ are of opposite sign, the quantity in the above square bracket is $(-1)^{K-1}K$. Thus, for any integer $K \geq 1$,

$$R_{L,K}'(-1) = \frac{(-1)^{K-1}}{2}\,K. \tag{A.23}$$

At the boundary $\xi = 1$, by (A.17) and (A.7),

$$R_{L,K}'(1) = \frac{1}{2}[P_K'(1) + P_{K-1}'(1)] = \frac{1}{2}[K(K+1) + K(K-1)].$$

Or, for any integer

$$R_{L,K}'(1) = \frac{K^2}{2}. \tag{A.24}$$

D. Lobatto Quadratures and Polynomials

Let $K \geq 2$ be an integer and f a smooth function on $I = [-1,1]$. Consider a quadrature formula that includes both boundary points:

$$\int_{-1}^{1} f(\xi) d\xi \approx \sum_{k=1}^{K} w_k f(\xi_k) \quad \text{where} \quad \xi_1 = -1 \text{ and } \xi_K = 1, \tag{A.25}$$

while all other ξ_k are parameters (unknowns); all w_k, $k = 1,...,K$ are also parameters. The total number of unknowns is therefore $2K-2$. To solve for these unknowns, we need $2K-2$ equations, which can be obtained by setting $f = \xi^k$, $k = 0,1,...,2K-3$, respectively, in the above. This system of equations yields all w_k's and $K-2$ values ξ_k where $k = 2,...,K-1$. These ξ_k together with $\xi_1 = -1$ and $\xi_K = 1$ are called the Lobatto points; the corresponding quadrature, Lobatto quadrature.

With K evaluations (including those at the two boundaries), the Lobatto quadrature has a degree of precision of $2K-3$ (as opposed to $2K-2$ of the Radau and $2K-1$ of the Gauss quadratures), i.e., it is exact for any polynomial of degree $2K-3$ or less.

Next, we derive the Lobatto polynomials. Consider the Legendre polynomials P_K and P_{K-2}. They are perpendicular to $\boldsymbol{P}_{K-3}$, i.e., $\mathrm{Pr}_{K-3}(P_K) = \mathrm{Pr}_{K-3}(P_{K-2}) = 0$. Set

$$\mathrm{Lo}_K = P_K - P_{K-2}. \tag{A.26}$$

Then Lo_K is also perpendicular to $\boldsymbol{P}_{K-3}$. In addition,

$$\mathrm{Lo}_K(-1) = \mathrm{Lo}_K(1) = 0. \tag{A.27}$$

The zeros of the above Lobatto polynomial are the Lobatto points. Denoting them by ξ_k, $k = 1,...,K$ where $\xi_1 = -1$ and $\xi_K = 1$, then the weights w_k's for the corresponding quadrature are

$$w_k = \begin{cases} \dfrac{2}{K(K-1)} \dfrac{1}{[P_{K-1}(\xi_k)]^2} & \text{if} \quad 1 < k < K, \\ \dfrac{2}{K(K-1)} & \text{if} \quad k = 1 \text{ or } k = K. \end{cases} \tag{A.28}$$

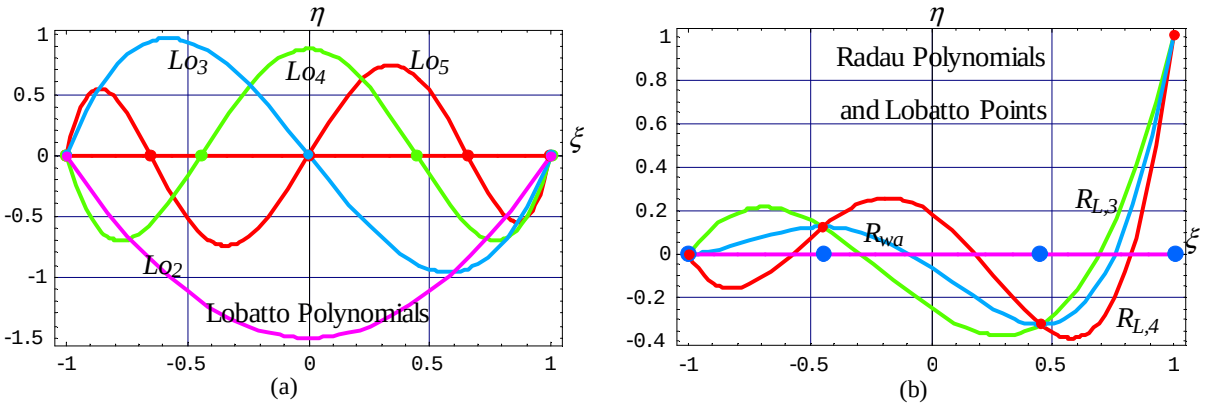


Figure A.3 (a) **Lobatto Polynomials on** $I = [-1,1]$. (b) **Polynomial** R_{wa} **and the Radau Polynomials.** *The polynomial* R_{wa} *reaches extrema at the Lobatto points (big dots) except at the right boundary; it is the correction function* $g_{\mathrm{Lump,\,Lo}}$ *for the right boundary. Note the different scales in* ξ *and* η *axes.*

The Lobatto polynomial can also be written as

$$\mathrm{Lo}_K(\xi) = \frac{-[K+(K-1)]}{K(K-1)}(1-\xi^2)P_{K-1}'. \tag{A.29}$$

Moreover, it can be expressed in terms of the Radau polynomials: for $K \geq 2$,

$$\mathrm{Lo}_K = 2(R_{L,K} - R_{L,K-1}) = 2\,(-1)^K (R_{L,K} - R_{L,K-1}). \tag{A.30}$$

Similar to the case of the Gauss and Radau points, the Lobatto points defined by the quadrature formula (A.25) are identical to the zeros of the Lobatto polynomials defined by (A.26). They are also given by ξ-coordinates of the intersections of the graphs of $R_{L,K}$ and $R_{L,K-1}$ as shown in Fig. A.3(b) (or those of $R_{R,K}$ and $R_{R,K-1}$).

E. Radau Polynomials, Lobatto Points, and Lumping.

Let $K \geq 2$ be an integer. To avoid carrying the factor $(-1)^K$, consider the left Radau polynomials $R_{L,K}$ and $R_{L,K-1}$. Their graphs intersect at K points whose ξ-coordinates are the Lobatto points, which are the big dots in Fig. A.3 (b). Let R_{wa} be the weighted average

$$R_{\text{wa}} = aR_{L,K} + (1-a)R_{L,K-1} \tag{A.31}$$

where a is chosen so that $\xi = -1$ is a double zero of R_{wa}, i.e., $R_{\text{wa}}'(-1) = 0$. To calculate a, by (A.23),

$$R_{L,K}'(-1) = \frac{(-1)^{K-1}}{2} K \qquad \text{and} \qquad R_{L,K-1}'(-1) = \frac{(-1)^{K-2}}{2} (K-1). \tag{A.32}$$

The above two slopes are of opposite sign; as a result, for $R_{\text{wa}}'(-1)$ to be 0,

$$a = \frac{K-1}{2K-1}. \tag{A.33}$$

That is,

$$R_{\text{wa}} = \frac{K-1}{2K-1} R_{L,K} + \frac{K}{2K-1} R_{L,K-1}. \tag{A.34}$$

Note that whereas $R_{\text{wa}}'(-1) = 0$, at the other boundary, a straightforward calculation using (A.24) yields

$$R_{\text{wa}}'(1) = \frac{K-1}{2K-1} R_{L,K}'(1) + \frac{K}{2K-1} R_{L,K-1}'(1) = \frac{(K-1)K^2 + K(K-1)^2}{2(2K-1)}.$$

That is,

$$R_{\text{wa}}'(1) = \frac{1}{2} K(K-1). \tag{A.35}$$

Note that this quantity is exactly the inverse of the Lobatto weight at the boundaries given by Eq. (A.28).

We claim that R_{wa} has extrema at all K Lobatto points except the right boundary. To prove this claim, it suffices to show that R_{wa}' vanishes at all Lobatto points except $\xi = 1$. Since R_{wa} is a polynomial of degree K, $(1-\xi) R_{\text{wa}}'$ is also of degree K. It suffices then to show that $(1-\xi) R_{\text{wa}}'$ vanishes at all Lobatto points. By definition (A.26) for the Lobatto polynomials, the problem reduces to showing that

$$(1-\xi) R_{\text{wa}}' = \text{const}\,(P_K - P_{K-2}). \tag{A.36}$$

where 'const' stands for some constant. Or, equivalently,

$$(1-\xi^2) R_{\text{wa}}' = \text{const}\,(1+\xi)(P_K - P_{K-2}). \tag{A.37}$$

To show that the above equation holds, first, expand R_{wa} in terms of the Legendre polynomials: by (A.34) and (A.17),

$$\begin{aligned} R_{\text{wa}} &= \frac{K-1}{2K-1} \left(\frac{1}{2}\right)(P_K + P_{K-1}) + \frac{K}{2K-1}\left(\frac{1}{2}\right)(P_{K-1} + P_{K-2}) \\ &= \frac{1}{2(2K-1)} [\,(K-1)P_K + (2K-1)P_{K-1} + P_{K-2}\,]. \end{aligned} \tag{A.38}$$

Next, denote by Q the quantity in the above square bracket:

$$Q = (K-1)P_K + (2K-1)P_{K-1} + K P_{K-2}. \tag{A.39}$$

Then, by differentiating Q and then multiplying by $(1-\xi^2)$, we have

$$(1-\xi^2) Q' = (K-1)(1-\xi^2) P_K' + (2K-1)(1-\xi^2) P_{K-1}' + K(1-\xi^2) P_{K-2}'. \tag{A.40}$$

We now rewrite the above in terms of P_k in two different expressions. For the first expression, by applying (A.9) to the first term on the right hand side of (A.40) and (A.10) to the last two, we obtain

$$(1-\xi^2)Q' = (K-1)\left[-K\xi P_K + K P_{K-1}\right] + (2K-1)\left[K\xi P_{K-1} - K P_K\right] + K\left[(K-1)\xi P_{K-2} - (K-1)P_{K-1}\right].$$

That is, after rearranging and observing that the term $K(K-1)P_{K-1}$ drops out,

$$(1-\xi^2)Q' = K\left\{-(K-1)\xi P_K - (2K-1)P_K + (2K-1)\xi P_{K-1} + (K-1)\xi P_{K-2}\right\}. \quad \text{(A.41)}$$

For the second expression, by applying (A.9) to the first two terms on the right hand side of (A.40) and (A.10) to the last term, we obtain

$$(1-\xi^2)Q' = (K-1)\left[-K\xi P_K + K P_{K-1}\right] + (2K-1)\left[-(K-1)\xi P_{K-1} + (K-1)P_{K-2}\right] + K\left[(K-1)\xi P_{K-2} - (K-1)P_{K-1}\right].$$

That is, after rearranging and observing again that the term $K(K-1)P_{K-1}$ drops out,

$$(1-\xi^2)Q' = (K-1)\left\{-K\xi P_K - (2K-1)\xi P_{K-1} + (2K-1)P_{K-2} + K\xi P_{K-2}\right\}. \quad \text{(A.42)}$$

By adding the quantities in the curly brackets in (A.42) and (A.41), the term $(2K-1)\xi P_{K-1}$ drops out. Thus,

$$\left[\frac{1}{K} + \frac{1}{K-1}\right](1-\xi^2)Q' = -(2K-1)(\xi+1)P_K + (2K-1)(\xi+1)P_{K-2} \quad \text{(A.43)}$$

The above simplifies to

$$\frac{1}{K(K-1)}(1-\xi)Q' = -(P_K - P_{K-2}) \quad \text{(A.44)}$$

Equations (A.38) and (A.39) imply $R_{\text{wa}} = Q/[2(2K-1)]$. By differentiating and then multiplying by $(1-\xi)$, we have $(1-\xi)R_{\text{wa}}' = (1-\xi)Q'/[2(2K-1)]$. The above then implies

$$(1-\xi)R_{\text{wa}}' = -\frac{K(K-1)}{2(2K-1)}(P_K - P_{K-2}) \quad \text{(A.45)}$$

Therefore, R_{wa} has extrema at all K Lobatto points except the right boundary. This completes the proof.

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